



Design of an Extended Handle for a Laser Scanning Tool

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Abstract:

This project aims to design a non-invasive extension handle for the Hexagon AS1 scanner. The scanner weighs 2.14 kg which exceeds the CCOHS recommended limit of 1.4 kg for one-handed tools and can lead to severe joint stress and inefficiencies when used overhead for prolonged periods as reported by MES engineers. The extension was intended to increase the scanning reach and allow remote actuation of the scanner's trigger while mitigating the ergonomic drawbacks of the existing configuration. An initial three iterations of the design were produced which saw major changes to the structure and material selection. A custom Python program was developed to validate the structural shaft selection under worst-case loading assumptions before they were procured. An additional three iterations resulted from finite element analysis. The final design features a composite carbon fibre main shaft for strength-to-weight ratio optimisation, machined aluminium for structural components, a custom Bowden cable operated trigger linkage and partially completed trigger locking mechanism. Physical deflection testing data of the carbon fibre shaft led to predicted deflection of 35.0 mm under a moment of 14.6 Nm. Finite element analysis simulated a 36.6 mm shaft deflection under the same moment. The assembled prototype demonstrated good functionality of the counterweight and trigger linkage systems, however high clamping forces at the custom joint adaptors caused them to crack, prompting the need for future iterations to focus on redesigning the attachment method for the shafts. Further development time should be spent on solderless cable attachment methods, weight reduction and fastener selection.

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1 INTRODUCTION

Metrology Engineering Services (MES) is an engineering team based in the UK which mainly focuses on delivering advanced, high precision measurement and damage assessment services in the aerospace sector, however, they have branched out into multiple industries where their skillset can also be applied. Part of their services are focused on laser scanning to capture the geometry of objects, from small scale such as a pneumatic actuator, to very large scale such as an entire air liner. The geometry capture is achieved using the AS1 Laser scanner developed by Hexagon. The AS1 can directly translate a physical object into a 3D model over a large area due to its tracking features which allows large objects, such as aircraft, to be scanned into a single model. This project focused on the AS1 in the configuration shown in figure 1.1 with the scanner attached to the AP21 handle to allow portable usage.



Figure 1.1: AS1 scanner module on the AP21 handle (Hexagon AB 2021).

An initial meeting took place at MES offices to define the project aims and the resulting requirements for the design. MES clearly laid out the project aims to design a non-invasive extension which would allow the scanner to reach previously inaccessible areas and alleviate the ergonomic challenges of using the AS1 for extended duration while also being stable and balanced enough to allow the user to accurately control the position of the scanner in space to maintain the required offset distance from the target, allowing the scanner to function correctly (figure 5.2). These features were especially desired for larger scale scanning work where the ergonomic drawbacks of the current configuration were more apparent.

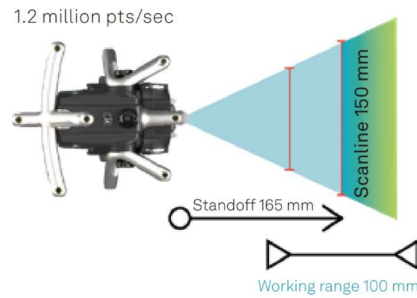


Figure 1.1: AS1 scanner range (Hexagon AB 2021) .

The aims directly established the project deliverables as:

Outputs:

- Delivery of a CAD model of the prototype at a minimum.
- A working prototype for testing with the AS1 if possible.

Outcomes:

- Faster completion times for large scanning projects.
- Less reliance on elevated platforms or ladders, creating a safer working environment.

Benefits:

- Quality of life improvements for the MES team using the extension through ergonomic enhancements.
- Financial savings for MES from reduced project cost due to faster completion times.

With the project clearly defined, a Gantt chart was produced (see appendix E) to plan the overall project timeline while the research phase began.

2 MARKET RESEARCH AND ERGONOMICS

2.1 Market Research

The first project action taken after meeting with MES was to conduct research into existing products available on the market which align with the requirements of this project. The criteria considered were a non-invasive attachment method, the extension of the reach of a hand tool and ergonomic improvement of an existing device. Two extension tools from Makita were considered. They offered extension handles for the DFR550 Auto Feed Screwdriver (figure 2.1) and TR001G 40V Max XGT Cordless Rebar Tying Tool (figure 2.2). Both featured a clamp-style attachment perpendicular to the devices grip and a trigger linkage to actuate the devices existing trigger. The auto feed screwdriver extension mainly appeared to enhance the reach of the existing device, with limited ergonomic improvement as the device would be front heavy. Alternatively, the rebar

tying tool extensions appeared to be more focused on ergonomics since it prevents the user from having to bend down to use the tool and it also features a support brace on its lower shaft.



Figure 2.1: DFR550 Auto Feed Screwdriver extension handle (Makita 2026a)



Figure 2.1: TR001G 40V Max XGT Cordless Rebar Tying Tool extension handle (Makita 2026b)

Another existing example is the Stihl GTA 26 Extension Shaft (figure 2.3), designed to provide a 1.5 m reach extension to their existing handheld mini chainsaw product. The saw attaches by fitting into a recessed housing and is secured using a toolless latch mechanism as shown in figure 2.4. Notably the GTA 26 has a low weight of 2.7 kg, which would make the device easy to use when extended at arms length. The weight distribution of the device would likely be towards the attached saw which would make the device front-heavy. This was a key consideration for the overall ergonomics of the design inspired from these existing solutions.



Figure 2.3: *Stihl GTA 26 Extension shaft (Stihl 2026)*



Figure 2.4: *Stihl GTA 26 attachment mechanism (Stihl 2026)*

After researching these extensions, the project focus moved to taking the features from each which seemed advantageous to use with the AS1 scanner and combining this with mitigations for the general ergonomic challenges inherent with long extension tools to generate some early concepts.

2.2 Ergonomics

The typical use case of the AS1 scanner attached to the AP21 by MES involves a large degree of overhead work and raised platforms to get the scanner close to its target. Figure 2.5 shows an example of this scenario with a worker using an AS1 on the AP21. Albers and Estill (2007) explain the potentially serious implications of long durations of overhead work on joint health and cite tool extensions as a mitigation method to allow devices to be used at elbow to hip level while greatly reducing joint stress levels. Fewster and Dickerson (2021) also qualify the risks of overhead work in its potential for causing musculoskeletal disorders and suggest that tools used for overhead work should have a mass lower than 1.25 kg.

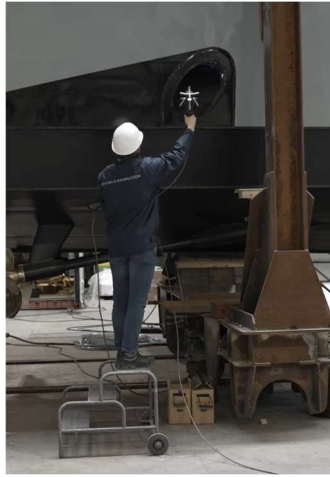


Figure 2.5: A worker holding an AS1-XL scanner overhead to scan a ship's hull (Hexagon 2025)

A further ergonomic disadvantage of the existing system is its high weight for a one-handed tool of 2.14 kg (Hexagon 2025) and its high centre of gravity relative to the user's gripping hand. The maximum recommended weight for a one-handed tool is 1.4 kg, and its centre of gravity should be aligned with the centre of the gripping hand according to the Canadian Centre for Occupational Health and safety (CCOHS) (2023). These guidelines also state that tools "above these recommendations should be supported with a counterbalance, tool holder, articulating arm, or straps as appropriate". This aligns with the discomfort indicated by MES in using the device as the device's centre of gravity is clearly not compliant with these recommendations. From this information, some key ergonomic features for the design were identified for the concepts. Since the weight of the scanner alone was relatively high and attaching it to an extension would only increase the required force from the user to keep the device stable, a counterweight would be required. A shoulder sling or harness was also determined to be a necessary feature to aid stability and shift some of the load from the user's arms to larger muscles in the torso. There are ISO and ANSI standards with relevance to this application, mainly concerning forestry work. They could not be accessed for this project so the CCOHS recommendation of having the centre of gravity of the device as close as possible to the gripping hand was used as a guideline to calculate the required counterweight mass to balance the device. Until the fourth iteration of the design, the gripping hand was assumed to be the rear hand which would also operate the trigger. This was later realised to be incorrect for a polearm style tool, where the forward hand is the main support hand. Therefore, the balance point was adjusted to be closer to the approximate position of the forward hand.

3 DESIGN METHODOLOGY

3.1 Technical Requirements

The first stage of the design process was to determine the requirements of the design. MES first indicated the challenges involved in using the AS1 scanner (see section 2.2). This allowed the project aims to be defined which led to five key technical requirements and an additional three optional desirable features. Tables 3.1 and 3.2 detail the key requirements and additional desired features respectively.

Table 3.1: Key requirements specified by MES

Requirement	Description
1. Extension arm	Increase the physical reach of the scanner device using a non-invasive attachment method
2. Trigger linkage	Trigger on the new device to actuate the devices existing trigger
3. Counterweight	Allow the device to be balanced and easy to control
4. Shoulder sling	Support the weight of the device to give more stability
5. Safety lanyard	Lanyard attached to AS1 to prevent it hitting the ground if it comes loose

Table 3.2: Additional desirable features specified by MES

Feature	Description
1. Physical protection of scanner cable connector	The connectors are fragile and are frequently damaged, so they require protection
2. Cable management/passthrough	Route the power and signal cable through or around the device to prevent it hanging from the end of the device
3. Repeater for indicator light	Allow the user to see the colour of the indicator light when it becomes obscured

3.2 Concept Generation

The initial concept generation was focused on iterating on requirement 2 to determine a non-invasive attachment method to be carried forward into the main concept. Two initial concepts for the attachment were inspired by the Stihl GTA 26 (Figure 3.1) and the Makita extensions (Figure 3.2). Both featured a Bowden cable operated trigger actuator, an adjustable joint, an attachment which conformed to the shape of the handle to secure it, and some protection for the attachment point.

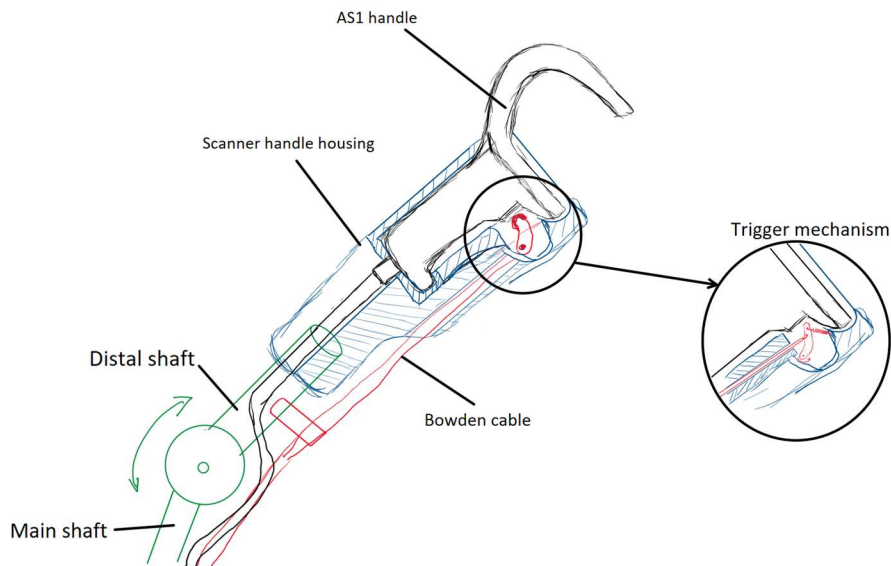


Figure 3.1: Stihl GTA 26 inspired concept

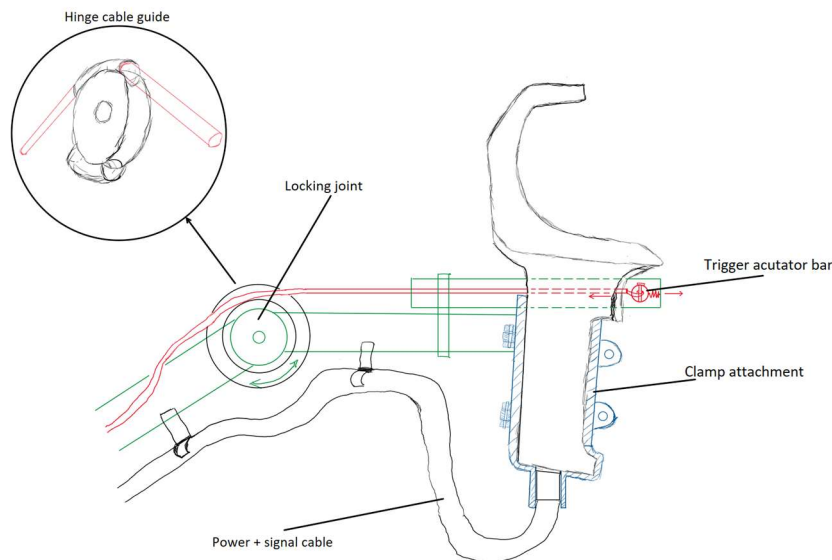


Figure 3.2: Concept inspired from the Makita devices.

Other methods of trigger actuation considered were electromechanical, and hydraulic. Servos could have been used to actuate the trigger with a wired connection and microcontroller. A simple hydraulic system could have been used for a low-slack, responsive trigger. The Bowden cable was chosen for its high mechanical efficiency (Chen et al. 2014), flexibility and lack of requirement for an additional power source as with servos. Bowden cables also have an advantage over hydraulics in reliability since the possibility of leaks introduces multiple potential points of failure, which would also be highly undesirable for the end user, in addition to potentially messy setup and

maintenance processes. Of these two concepts, the Stihl GTA 26 inspired attachment was chosen to carry forward for the final concept. Primarily due to the increased protection for the scanner cable connection since there would be more material surrounding the connection. It would also reduce the likelihood of the cable snagging since the cable would be axially aligned with the distal shaft and so could run over or inside the shaft. The range of motion and possible scanning arc for this concept also appeared more practical for use on a pole arm style tool since it would allow scanning over the top and behind some objects below the height of the scanner which the other concept would be less suited to assuming a joint range of motion of ± 90 degrees from the main shaft axis. Figure 3.3 shows the initial concept for the handle, counterweight and extension shaft design. It featured a bike brake or clutch lever to actuate the Bowden cable and a rough concept for a trigger locking mechanism to prevent finger strain from continuously pulling on the trigger.

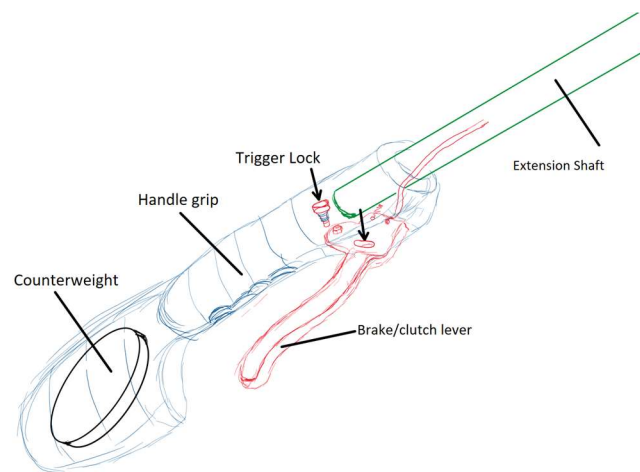


Figure 3.3: Initial handle and counterweight concept

Another concept sketch was created to define the attachment mechanism in more detail. Taking further inspiration from the Stihl example, it featured a toolless latch mechanism. Figure 3.4 shows the concept and the order of operations to close the latch. The concept also added the attachment point for a safety lanyard on the distal shaft.

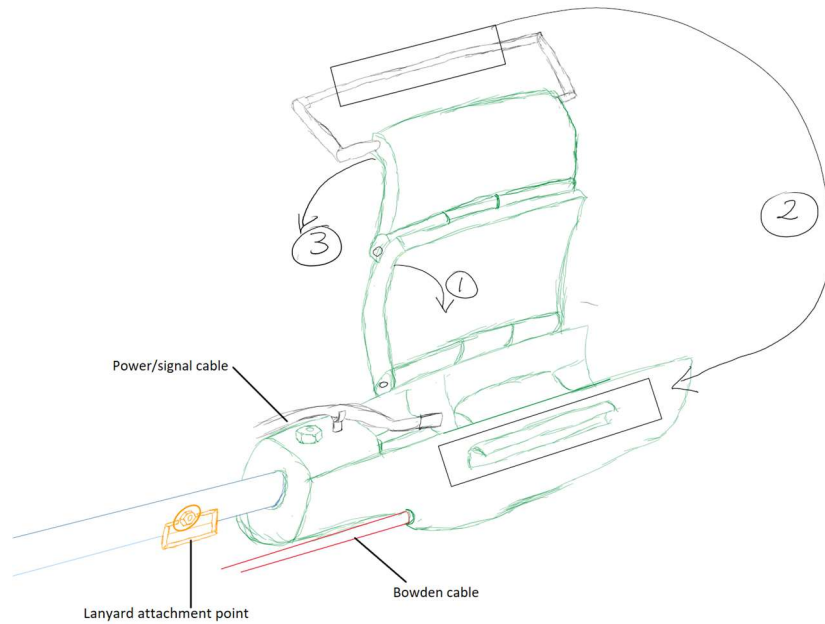


Figure 3.4: Latch mechanism concept

The concepts shown were then combined and refined into a full concept to be presented to MES for approval before CAD development started. For ease of viewing, the concept is split into 3 figures. Figure 3.5 shows the counterweight and handle, figure 3.6 shows the main shaft and joint, and figure 3.7 shows the distal shaft and scanner attachment. See the appendix A for the full concept image.

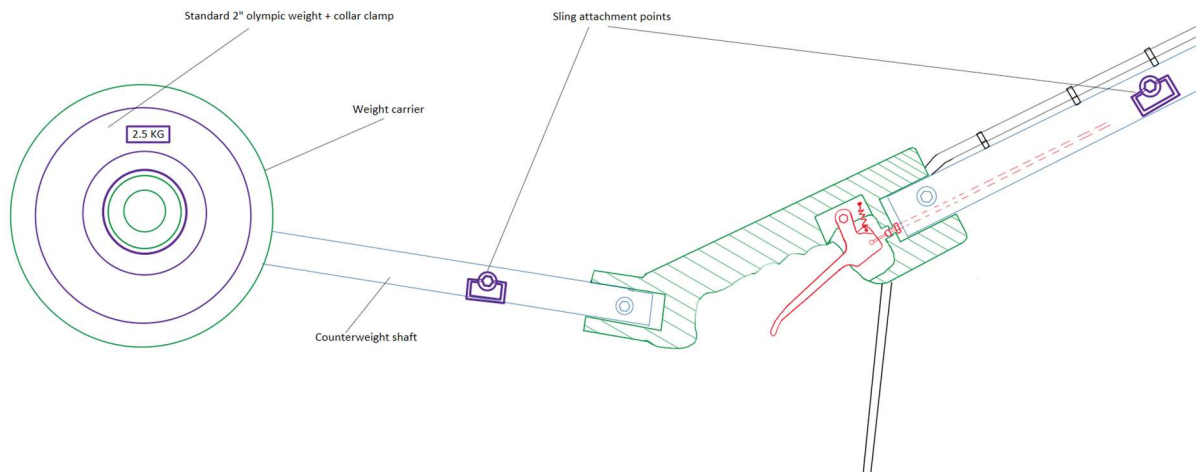


Figure 3.5: Counterweight and handle concept

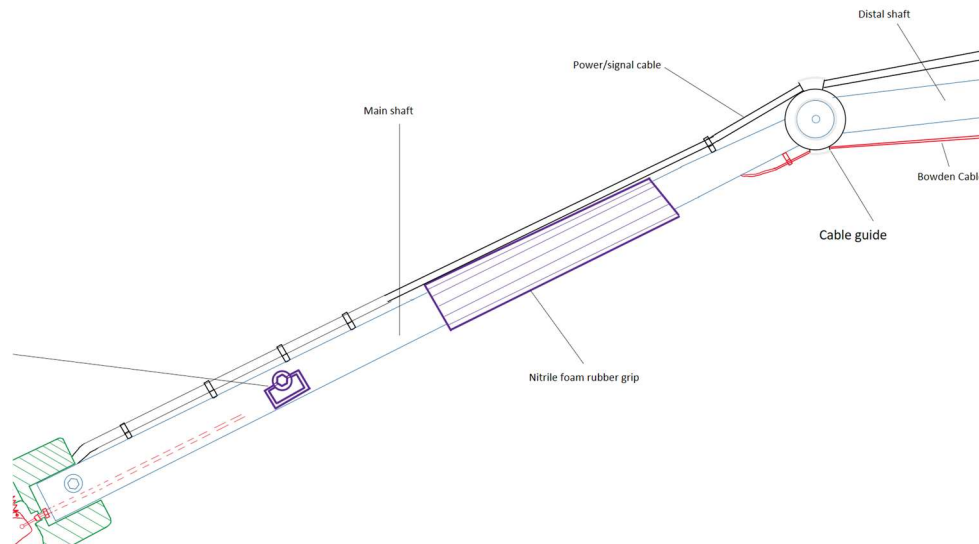


Figure 3.6: Main shaft and joint concept

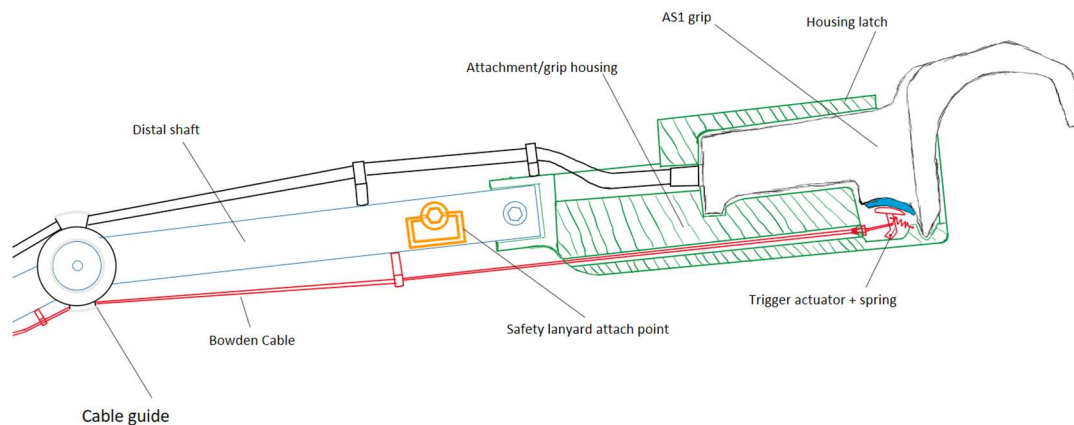


Figure 3.7: Distal shaft and scanner grip housing concept

The full concept was presented to MES and was overall approved and their feedback was used to inform some design changes before the CAD modelling phase began. In addition to this concept, a small plastic fibre optic cable designed for display lighting was procured and demonstrated to be able to pick up led light at one end and relay it to the other (see figure 3.8). This was presented as a method to implement feature 3. Some questions were also posed to MES to clarify some details regarding how the extension would be used. A summary of the meeting and the resulting design alterations is shown in table 3.3.

Table 3.3: Concept presentation meeting summary

Question	Answer	Action
How frequently will the AS1 be removed from the extension?	Not very often, most likely at the start and end of the day.	Complex toolless hinge design may be unnecessary. Consider using a simpler bolted hatch to secure the AS1.
Would a trigger locking mechanism be beneficial?	Yes, it is already an ergonomic concern for the AS1 in its current state.	Carry the locking mechanism concept forward into the next design phase.

Would the power/signal cable be required to be fully enclosed inside the main shaft?	Not necessary, cable management loops would be enough.	Explore designs to attach cable management clips to the main shaft.
Feedback	Answer	Action
Distal shaft length	The distal shaft is too long, it would be preferred if the housing attaches directly to the joint, or if the shaft is as short as possible.	Shorten the length of the extension distal from the joint.
Shaft insertion stresses	The depth of the shaft insertions into the handle and housing appears very short and could cause high stress concentrations.	FEA is already planned as part of the design process. Increase the depth of the shaft insertions for the CAD model.
Fibre optic demonstration	It is a “nice to have” feature, but not high priority or essential.	Use procured fibre optic cable in the prototype for demonstration and focus on overall design.

3.3 Material Selection

During the concept phase and first two iterations of the CAD model, the weight carrier, handle grip and scanner housing were intended to be manufactured using 3D printing in PLA. This method was selected since it would produce lightweight components and so reduce the overall weight of the extension. It also allowed these components to be designed without careful consideration of their machinability. After the second iteration (see section 3.5.2) was completed in CAD, finite element analysis (FEA) was started in Ansys. Complications with simulating the material of a 3D-printed component were immediately encountered. Perera et al. 2025 demonstrate how the mechanical properties of 3D-printed parts cannot be accurately modelled using standard material properties due to their internal cavities and layered structure. This study was limited to simple external specimen geometries and involved detailed modelling of different internal structures before subjecting them to simulated loads. It was determined that the process of modelling a specific internal structure across several components of more complex geometry would be an unacceptable expansion of scope for this project. However, an FEA study was needed to ensure that the extension would be strong enough to prevent damage to the high value AS1 scanner. The solution opted for was to take a hybrid approach of using a machinable, homogenous material for the key structural components to allow accurate simulations, and 3D-printing the non-structural components with more complex geometry which made them more difficult to machine. Initially, a high-strength, machinable polymer, acetal homopolymer (POM-H) was favoured to attempt to keep the weight to a minimum, however the procurement challenge and cost of obtaining large enough billets of POM-H was too high to machine more than one prototype version, so the first machined parts would have to be the final prototype. Since 3D-printed parts were determined to be unreliable for structural components, the structure was changed to use aluminium alloy shafts for the handle and counterweight shafts, machined

aluminium for other structural components and a carbon fibre shaft as the main shaft for its far higher strength to weight ratio compared to aluminium alloy.

3.4 Part Procurement

Initially, the intended design methodology was to design every component in the extension from scratch and have them custom made. The first critical component considered was the locking joint due to its complexity. Figure 3.8 shows an initial concept sketch of a potential locking joint design.

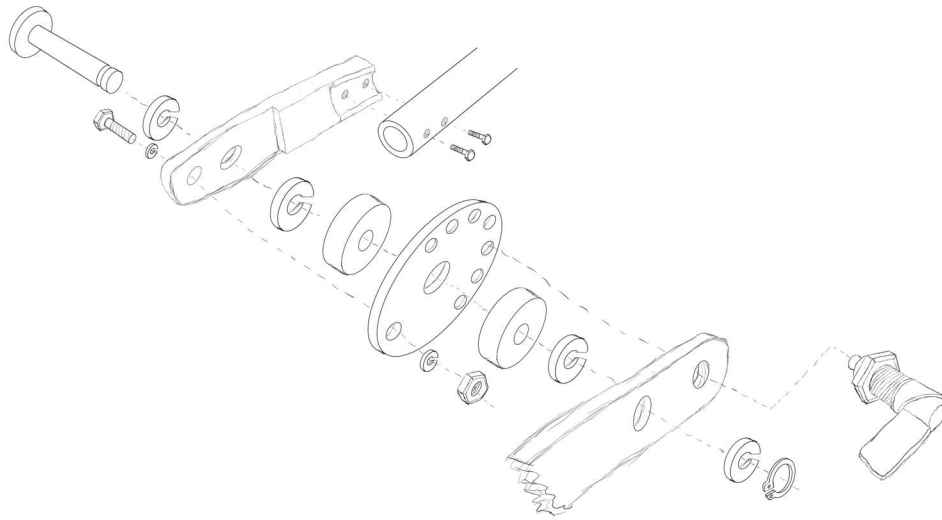


Figure 3.8: Initial locking joint concept.

After producing this sketch, it was determined that the design of a strong and functional locking joint would considerably expand the scope of the project and would likely prevent the project deliverables from being met. To speed up the development process and avoid unnecessary extra component design, a new procurement methodology was adopted to buy as many components off the shelf as possible and design the CAD model around these components. Table 3.4 shows the bill of materials list for off the shelf parts procured during this project.

Table 3.4: BoM table for off the shelf components.

Part	Description	Part/Stock N°	Supplier
Locking Joint	Square profile adjustable locking joint with 180-degree range of motion rated to withstand moments of 35 Nm to 100 Nm depending on position	267-3074	RS Components
Clutch lever	Basic low-cost cable operated clutch lever for proof of concept to actuate trigger linkage in prototype	[UPC] 714046494241	Amazon
Spring Plunger	Polymer construction M10 x 1 mm spring loaded plunger for trigger locking mechanism	918-5276	RS Components
Fibre optic	Low-cost fibre optic for display lighting for proof of concept for the indicator light repeater	RM7434949638667	Amazon
Carbon fibre tube	Main shaft of extension	CFT-WF-30-27-1	Easy Composites

Aluminium handle shaft	22.2 mm x 3.2 mm aluminium alloy tube for the handle shaft in v0.3	13368	Metals4U
Aluminium counterweight shaft	25.4 mm x 1.6 mm aluminium alloy tube for the counterweight shaft in v0.3	13370	Metals4U
D-ring cleat x 3	Attachment points for the shoulder strap and safety lanyard.	SSDRC0320	GS Products
Trigger spring	Compression spring to reset trigger actuator	751-641	RS Components
Bowden cable	Motorbike throttle cable repair kit	[MPN] UTCKIT	Deamon tweaks
Shoulder strap	Tool bag shoulder strap to support the extension.	474FU	Screwfix

Most components were delivered within a short time span of ordering; however, the locking joint was ordered in December 2025 and did not arrive until late February 2026 which resulted in some late-stage redesigning of the attachment points (see section 4.1.2). To approximately verify whether the selected dimensions of the shafts in the third iteration of the CAD model were adequate, the weights and resulting moments of each component about the desired centre of gravity were calculated in excel, and an optimisation program to determine the minimum required dimensions was written in python. Table 3.5 shows the moment calculations and table 3.6 shows the resulting balance calculations to find the required counterweight mass and approximations of the torque and bending moments on each shaft. The bending moments were taken along the shaft axes and the torques were calculated as the moments of the weight carrier and the combined weight of all the components distal from the joint about the shaft axis if the extension were held at 90 degrees to its normal orientation with the joint angle at 90 degrees.

Table 3.5: Moment calculations for v0.3

v0.3 Parts						
Part	Volume (mm ³)	Material	Density (kg/m ³)	Mass (kg)	Distance (m)	Moment (Nm)
Carrier	1874338.45	Al	2700	5.0607138	0.6	29.787362
Carrier shaft	59372.33	Al	2700	0.1603053	0.3	0.4717785
Handle Boss +join block	572221.72	Al	2700	1.5449986	0.01	0.1515644
Main Shaft	N/A	CF	N/A	0.1938	0.57	1.0836715
CF joint adaptor	45533.73	Al	2700	0.1229411	1.14	1.3748992
Joint (Approx)	131735.26	Diecast Zinc	7600	1.001188	1.19	11.687768
Al joint adaptor	48845.38	Al	2700	0.1318825	1.23	1.5913341
Housing shaft	14333.18	Al	2700	0.0386996	1.31	0.4973322
Housing ext	328005.12	Al	2700	0.8856138	1.6	13.900595
Housing int	335461.66	PLA	250	0.0838654	1.6	1.3163516
Housing Hatch	49485.91	Al	2700	0.133612	1.6	2.0971733
Scanner		N/A	N/A	2.14	1.66	34.849044
			Solid PLA: 1250			
			Assumed 20 % infill			

Table 3.6: Selected shaft dimensions and calculated loads

v0.3 Balance - Moments about handle grip					
Total Forward moment	Total Backwards moment	Balance moment required	Counterweight mass / kg		
68.5497331	30.25913999	38.29059311	6.505367501		
Shaft	Length / m	Selected thickness / mm	Selected OD / mm	Torque / Nm	Bending Moment / Nm
Carrier	0.5	1.6	25.4	13.47704751	68.5497331
Handle	0.3	3.2	22.2	20.35560092	68.5497331
Main	1	3	30	20.35560092	68.39816873
Housing	0.08	3.2	22.2	3.181472463	10.98320639
Sum of mass distal from joint:	4.414861284	43N			
Joint to scanner dist:	0.47				
Material properties	Tensile limit / Mpa	Shear limit / Mpa			
CF	650	30			
AL	340	210			
Total mass	18.00298761				
Sum of weight distal from handle shaft	61.5734558				

The bending moment results in Table 3.6 were multiplied by a safety factor of 2.5 and converted to an equivalent cantilever load assuming each shaft were rigidly supported at one end, and the load was applied to the other end. The torque results were also multiplied by the same safety factor. This method was used to give a worst-case loading scenario since the project timescale mandated that the shafts would have to be ordered before FEA could be carried out. The material properties for the aluminium shafts and the tensile modulus of the carbon fibre shaft were obtained from the suppliers' websites. The shear stress limit of the carbon fibre shaft was conservatively taken as 30 MPa, the lower limit found in (Bradley et al. 2007). The selected shaft dimensions were validated by fixing the outer radius (R_o) limits and unrestricting the inner radius (R_i) limits so they would be calculated by the program. The graphical output of the program for the carbon fibre shaft is shown in figure 3.9. As expected, the torsional limit was the limiting factor for the carbon fibre shaft. Table 3.7 shows the results of the program for each shaft. The full source code of the program is available in Appendix D.

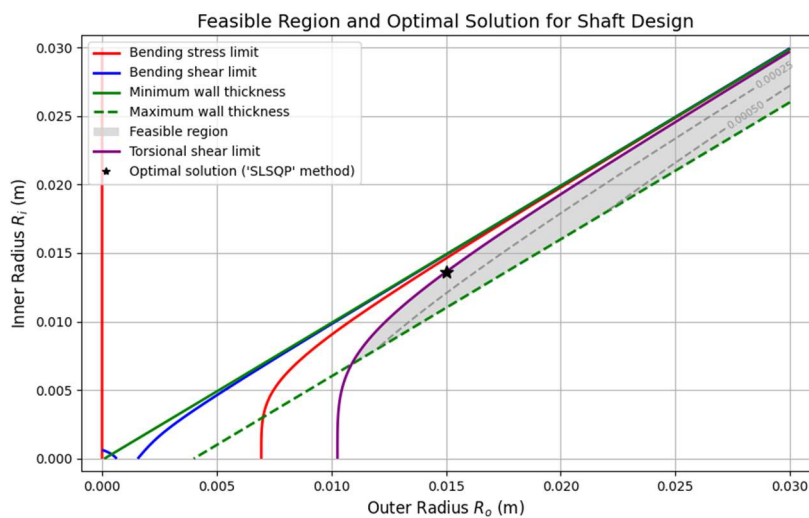


Figure 3.9: Graphical output of optimisation program for 30 mm outer diameter carbon fibre shaft.

Table 3.7: Optimisation results for each shaft.

Shaft	Torque x 2.5 (FoS) / Nm	BM x 2.5 (FoS) / Nm	Cantilever equivalent end load / N	Selected Ro / mm	Min Ri / mm	Min thickness / mm
Carrier	33.69261877	171.3743327	342.7486655	12.7	11.560019	1.139981
Handle	50.88900231	171.3743327	571.2477758	11.1	9.475154	1.624846
Main	50.88900231	170.9954218	170.9954218	15	13.617953	1.382047
Housing	7.953681158	27.45801598	343.2251998	11.1	10.885352	0.214648

The results validated that the selected shafts would be unlikely to fail catastrophically under these artificial conditions since the calculated minimum thickness was lower than the corresponding selected thickness of each shaft. Notably the program omitted the calculation of shaft deflection which could have been a useful metric to validate the program in comparison to FEA.

3.5 CAD Development

3.5.1 Baseline CAD Geometry from Concept

Figure 3.10 shows the first iteration (v0.1) of the CAD model generated from the concept including the AS1 geometry in the housing.

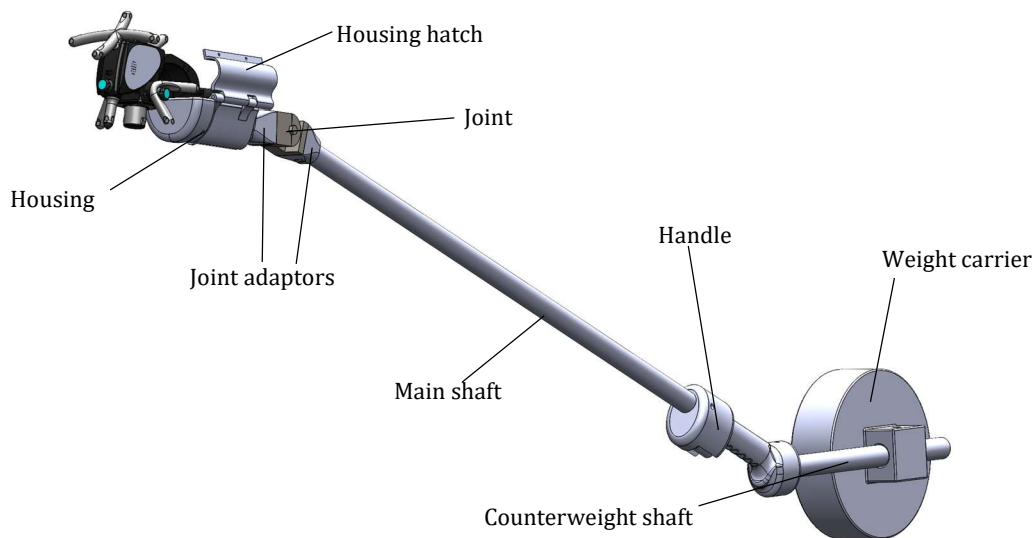


Figure 3.10: Initial CAD geometry from concept

This iteration outlined the model's overall structure and provided a base upon which to refine the design. The handle (figure 3.12) was modelled as a single piece of material with the addition of a shaft to attach the clutch lever clamp. The main shaft featureless and some required details were not modelled such as cable management for the power and Bowden cables and a trigger actuator for the existing AS1 trigger in the housing. An approximation of the selected locking joint was modelled to complete the assembly. To accommodate the joint, an adaptor shown in figure 3.11 was modelled to connect the main shaft to the joint.

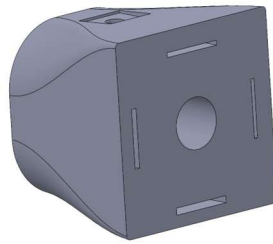


Figure 3.11: Main shaft to joint adaptor

Figure 3.13 shows the housing. It was modelled to conform to the geometry of the AS1 grip with a slight scale factor to allow a sliding fit. The housing connected directly to the joint to replace the distal shaft and reduce the length of the distal section. Following the feedback on from MES, the decision was made to simplify the design of the securing hatch to simply be bolted down to the housing using threaded heat inserts to be melted into the housing. The same securing method was intended to attach the housing and joint adaptor to the locking joint.

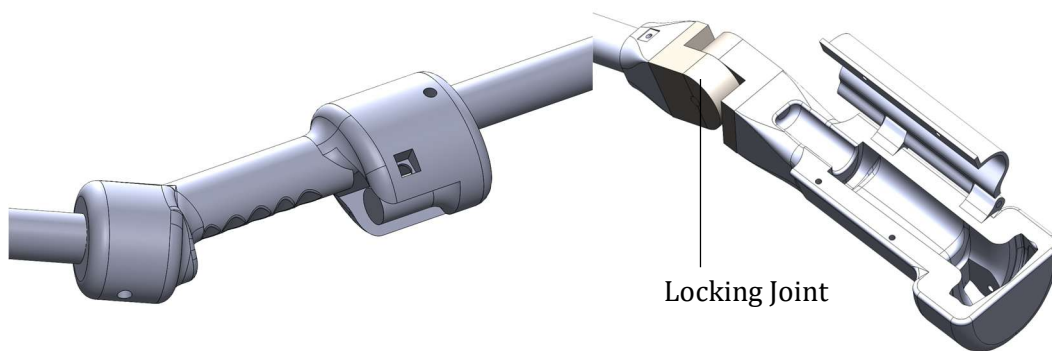


Figure 3.12: Initial CAD model of the handle

Figure 3.13: Initial CAD model of the housing3

After evaluating the first design, some clear issues were observed. Firstly, there was an oversight with the clutch lever attachment shaft. The clamp bolted on from the back so assembly would be difficult. In addition, considering the ergonomics of the attachment location, it would likely be too far away from the handle grip for a typical hand size to comfortably reach. The reliance on heat inserts for structural support was a concern since their strength could easily be lower or inconsistent if improperly installed, and it would be difficult to validate in FEA.

3.5.2 Initial Refinement Iteration

The second version of the model (v0.2) aimed to remedy the assembly issues and add unmodelled features while still being designed to have the same components 3D-printed as v0.1. The main fastening method was changed to bolting through components to increase their strength and reliability. The handle was redesigned by tapering the grip into an attachment shaft for the clutch lever clamp in line with the grip axis to allow it to be more easily reached. The material above this shaft was removed to allow assembly. A slot was cut into the main shaft to run the Bowden cable

inside it. Regularly spaced holes for cable management clips to attach using plastic pop-rivets and several holes for a D-ring to attach for the shoulder sling to clip onto were cut into the main shaft. A second D-ring attachment was also intended at the counterweight shaft bolt on the handle. Figures 3.14 and 3.15 show the handle and a cross-section through it respectively.

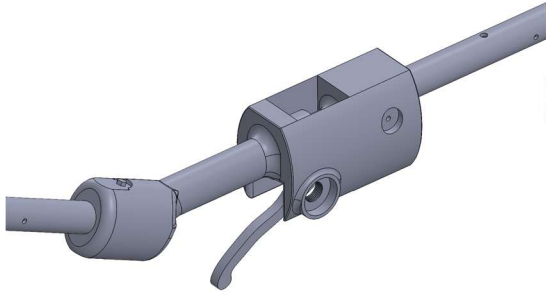


Figure 3.14: *Refinement iteration of handle*

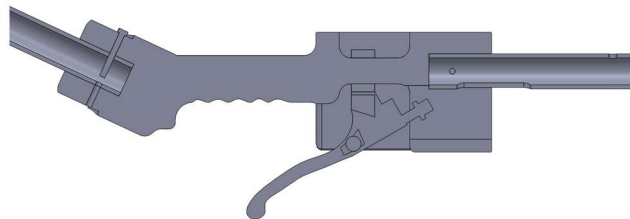


Figure 3.15: *Cross-section through v0.2 handle*

The weight carrier was redesigned to position the counterweight load directly below the counterweight shaft (figure 3.16) to prevent the handle from twisting out of the user's hand which was an additional ergonomic concern with v0.1.

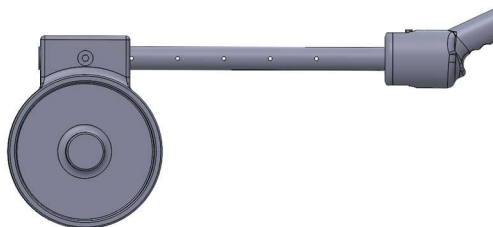


Figure 3.16: *v0.2 weight carrier redesign*

Figure 3.17 shows the second iteration of the housing assembly. The distal shaft was reintroduced to the design with a slot to allow the Bowden cable to be routed into the centre of the housing to reach the trigger actuator, which was also added to this iteration. An additional adaptor was added between the joint and the distal shaft. A cable guide funnel was added at the pivot point of the joint to maintain a consistent Bowden cable length along the distal section to help prevent slacking. Figures 3.18 shows a cross section of the housing assembly and figure 3.19 shows a close-up cross-section of the trigger actuator where the tension spring attachment points can be seen which would resist the pull from the Bowden cable to allow the actuator to reset when the clutch lever at the handle was released.

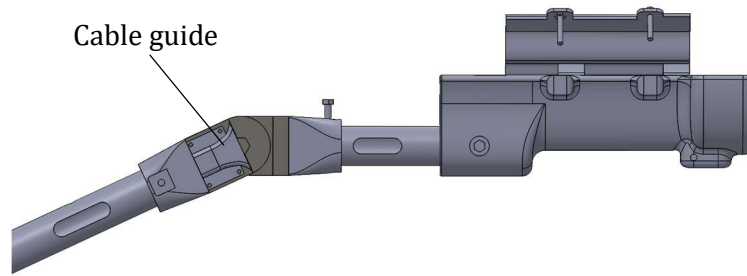


Figure 3.17: v0.2 housing assembly redesign

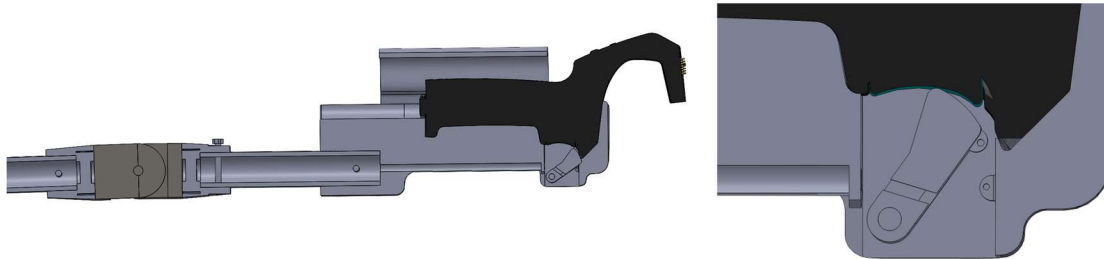


Figure 3.18: v0.2 housing assembly cross-section

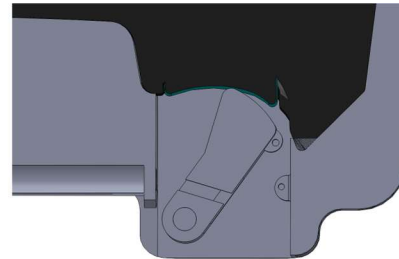


Figure 3.19: v0.2 Trigger actuator cross-section

3.5.3 Machinability and New Material Selection Redesign

After the change in material selection to aluminium, carbon fibre and 3D-printed PLA, the extension was redesigned to accommodate this approach. Each shaft attachment was secured with a split-hub clamping method to allow for rigid connections between the machined aluminium components and the shafts without relying tight tolerances and to mitigate edge wear at the contact point between the epoxy in the carbon fibre shaft and steel bolts. The profile of the extension (figure 3.20) was straightened in v0.3 to allow the counterweight shaft and handle shaft to fit together. Figure 3.21 shows a cross section through the handle with the counterweight shaft highlighted in blue. The handle from v0.2 was divided into separate components with the grip as a non-structural, 3D-printed part and the joining block and front handle boss were designed to be machined from aluminium alloy.



Figure 3.20: v0.3 Full assembly

Handle shaft

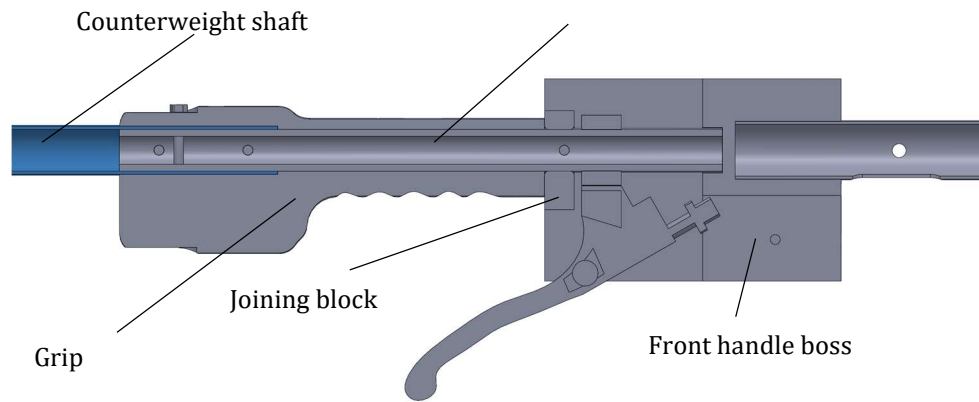


Figure 3.21: v0.3 Handle cross-section

Figure 3.22 shows the updated housing assembly in cross-section, with the trigger actuator resisted by a compression spring rather than a tension spring due to the limited clearance within the housing. The housing was separated into two components, and external aluminium alloy shell with a removeable 3D-printed seat for the AS1 grip to sit inside as shown in figure 3.23. 3D-printing the seat allowed the more complex and difficult to machine geometry surrounding the AS1 grip to be manufactured easily whilst the aluminium provides greater strength. The housing hatch was also intended to be machined out of aluminium alloy.

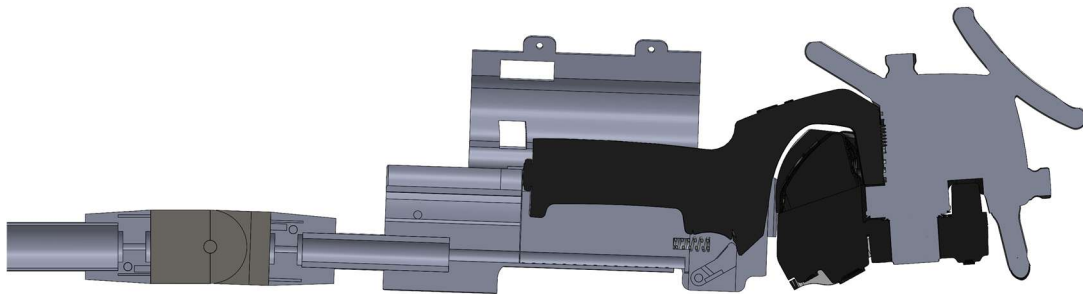


Figure 3.22: v0.3 Housing assembly cross-section.

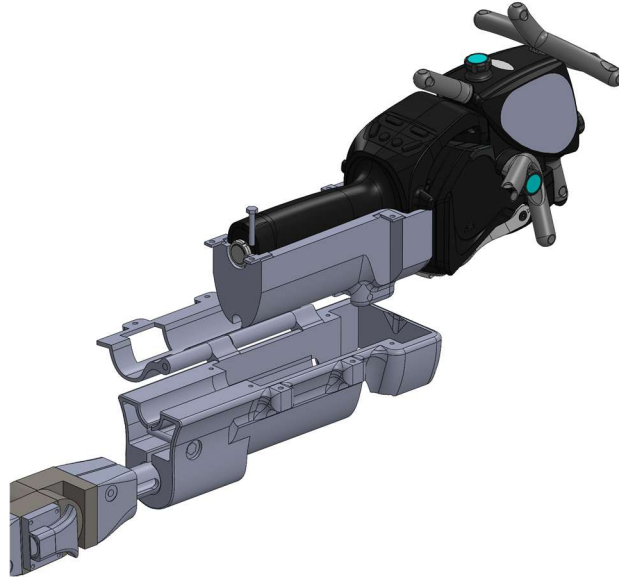


Figure 3.23: v0.3 Housing assembly with printed seat removed.

The material selection for the main shaft had now changed to a composite carbon fibre shaft to reduce the total weight of the extension by reducing the weight of the shaft itself and reducing the required mass of the counterweight. Further images of v0.3 can be found in appendix A. With v0.3 completed, FEA could begin to validate the strength of the design to ensure it would not fail and cause the scanner to be damaged.

4. VERIFICATION AND TESTING

4.1 Finite Element Analysis

4.1.1 Structural Shaft Analysis and Iterative Development

The initial FEA methodology used was to consider parts individually in Ansys to conserve element counts and memory in each simulation and apply theoretical calculated loads at the contact points in the assembly. The first section to be analysed was the handle shaft and joining block since it was the core structure of the design and would determine the overall structural integrity of the extension. The loads from the weight carrier and joining block were set as bearing loads calculated to balance the moments in table 3. The bearing loads were calculated with

$$F = M/d \quad (1)$$

where F is the bearing load force, M is the moment to be balanced, and d is the width of the cylindrical contact. For example, the 68.5 Nm forwards moment on the handle shaft, this would be applied to the 30 mm wide contact between the join block and the handle shaft with a three times safety factor to give a 6850 N bearing load as shown in figure 4.1. The vertical bolt at G which goes through the handle grip and the section of the shaft in contact with the grip were

modelled as cylindrical supports. The calculated torques were simulated as direct forces applied to opposite halves of the bolt holes to form couples. The distal end of the handle shaft where it sits in the front handle boss.

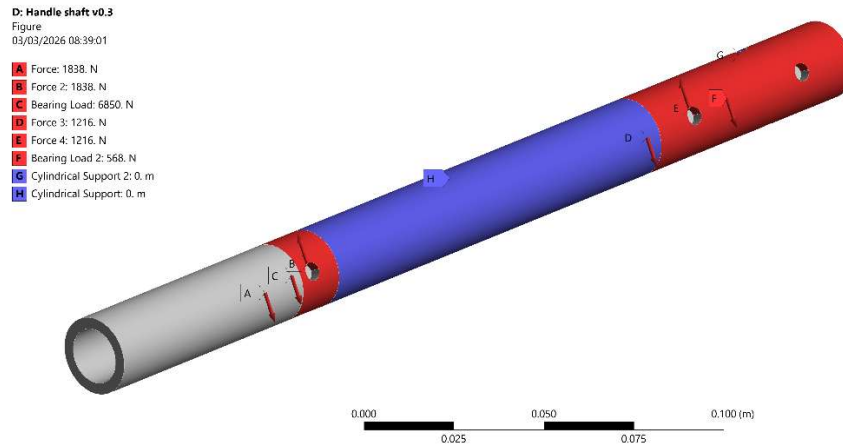


Figure 4.1: Initial handle shaft loading simulation.

Figure 4.2 shows the initial result at a $5e-3$ m mesh size with a peak stress of 202 MPa inside the join block bolt hole. There were no concerning stress points at the counterweight end of the shaft.

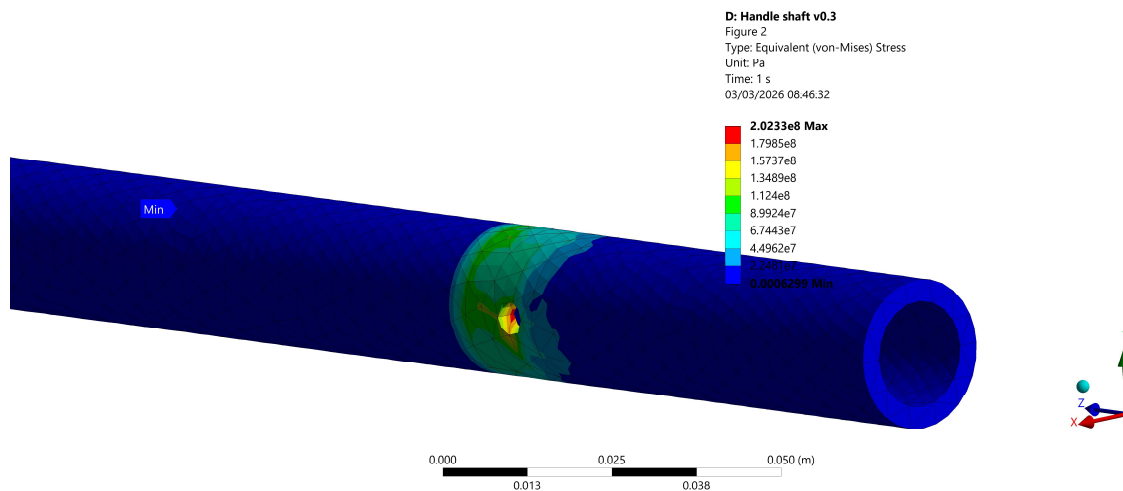


Figure 4.2: Initial v0.3 handle shaft result at $5e-3$ m mesh size.

After locally refining the mesh around the join block load, the maximum stress moved to the edge of the bearing load and increased significantly to 323 MPa. Due to the safety factor and 340 MPa yield strength of the selected shafts this stress line was not concerning. In addition, the stress rapidly reduced at small distances either side of the maximum point suggesting it was an artificial stress concentration caused by the boundary condition of the bearing load (figure 4.3).

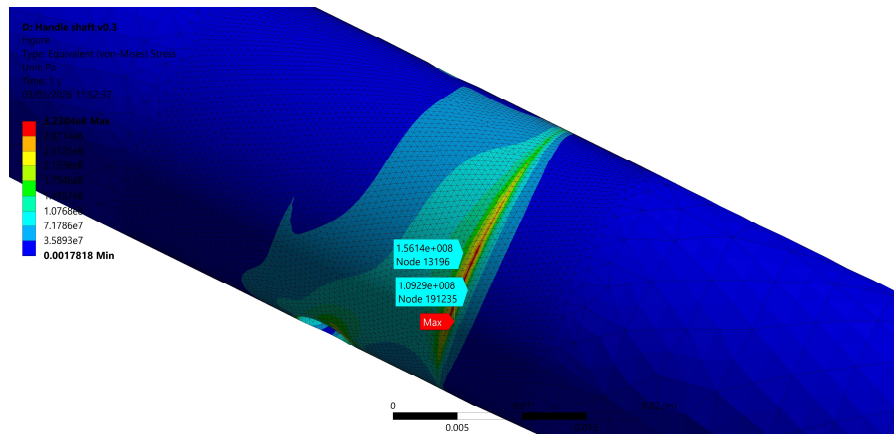


Figure 4.3: Local mesh refinement of initial setup for v0.3 handle shaft.

Consideration of the first approach's accuracy to the actual loads applied to it inspired a change to split the contact surface bearing loads onto the upper and lower halves of the cylinder to create couples to simulate the bending moments the shaft would experience (figure 4.4). Figures 4.5 and 4.6 show the results with a more pronounced stress concentration with a higher maximum of 542 MPa at the bearing load edge and inside the join block bolt hole. Although these small stress points seemed more likely to cause some local deformation and work hardening, they were still not considered highly realistic due to the arbitrary sizes of the contact zones being exactly half of the shaft's surface. It was determined that this approach would be difficult to make more accurate without carrying out Hertzian analysis on every contact which would be too time consuming in addition to the FEA.

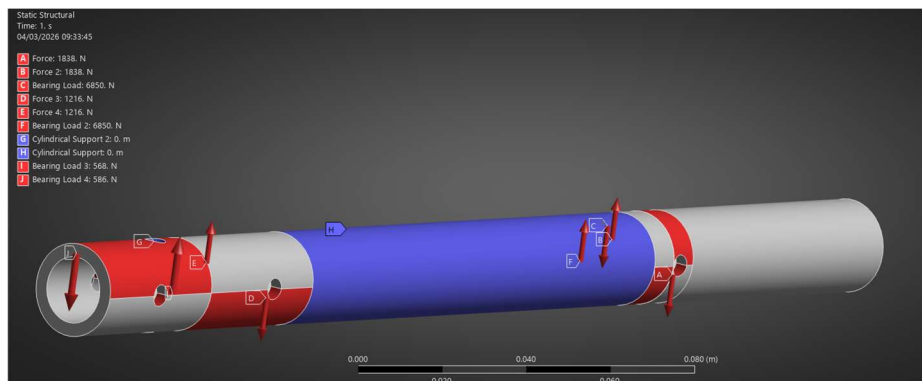


Figure 4.4: Revised bearing load setup for v0.3 handle shaft.

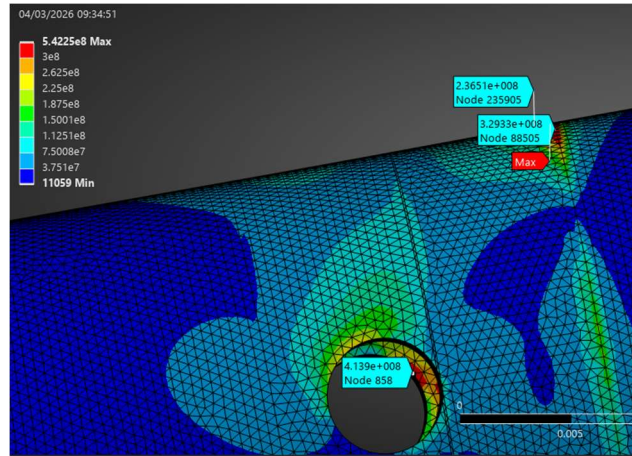


Figure 4.5: Revised bearing load results for v0.3 handle shaft.

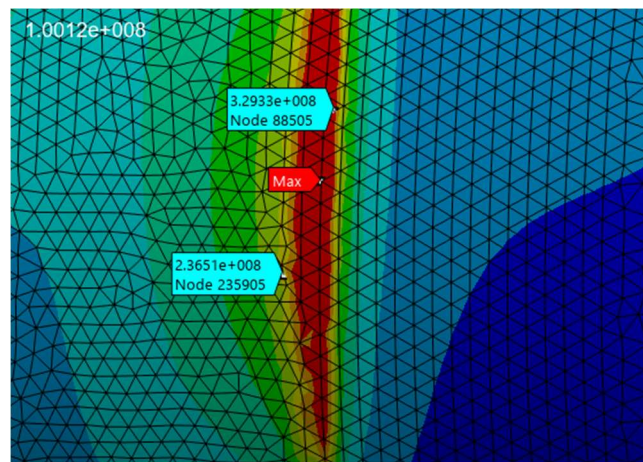


Figure 4.6: Revised bearing load boundary stress concentration for v0.3 handle shaft.

Due to low confidence in this approach, a final method was adopted for the handle shaft. A three-part assembly of the shaft, the join block and a bolt was created, with contacts set to no separation and the same cylindrical supports as previous approaches. A remote force equal to the counterweight was applied to the section supporting the counterweight shaft (D in figure 4.7). This remote force acted at the relative location of the weight carrier. An additional 68.5 N force was applied to the sections of the join block faces highlighted in figure 4.7. This force acted vertically downwards, parallel to the face at 1 m to simulate the previously calculated handle shaft moment. A 1 mm radius fillet was added to the internal edge of the join block during the assembly creation to mitigate artificial stress concentrations at the sharp circumferential edge. The torques applied directly to bolt holes in previous approaches were not included since, in the second approach, the bearing loads were suppressed and the maximum stress from the torques reduced to 223 MPa. Therefore, it was concluded that the main area of concern was the bending moments. In addition, including the torques with this setup was considered an unrealistic loading scenario since the torques would only be present if the extension were held on its side. The remote forces

in this approach were set as the actual calculated load values without a factor of safety built in to give a better gauge of the real stresses in each part.

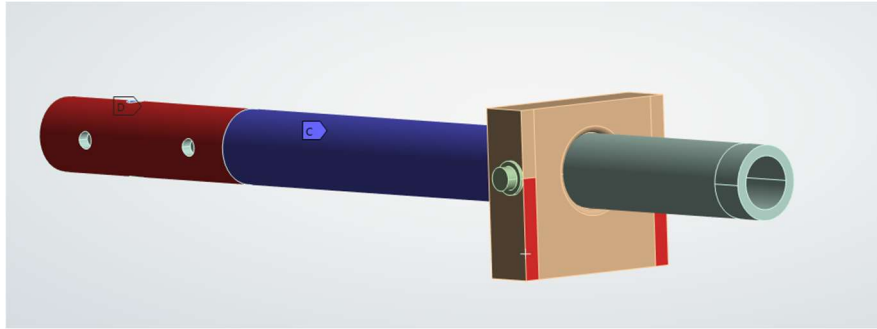


Figure 4.7: Final remote force approach setup for v0.3 handle shaft assembly.

Figure 4.8 shows the result with a maximum stress concentration of approximately 545 MPa at the join block bolt hole after a local mesh refinement around the joint block area to $5e-4$ m. This stress appeared to be a singularity and did not present any structural concern since the stress dropped off sharply to larger areas of around 150 MPa which would give a general factor of safety of 2.26. Figure 4.9 shows a similar result in the join block. The maximum deflection was recorded at 2 mm (see appendix B).

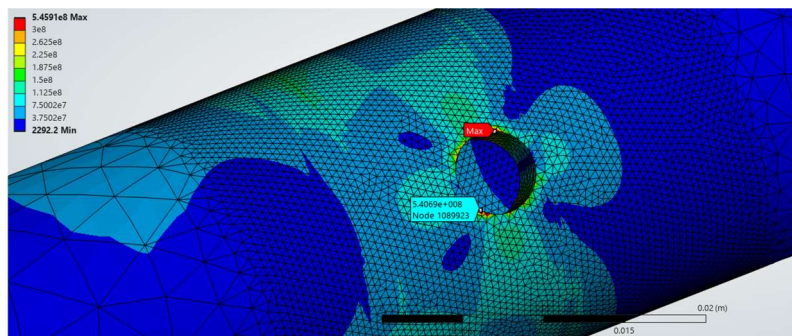


Figure 4.8: Final remote force approach handle shaft result for v0.3.

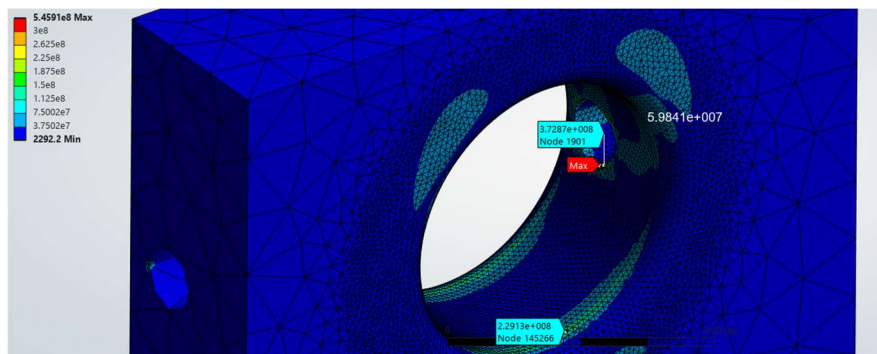


Figure 4.9: Final remote force approach joining block result for v0.3.

The next part of the design analysed was the main shaft and front handle boss. Since the main shaft was secured by clamping the boss around it, the required bolt force to resist the maximum shaft torque would need to be calculated. The clamping force was determined using

$$F_c = \frac{T \times SF}{\mu \times r} \quad (2)$$

where F_c is the required clamping force, T is the torque, SF is the safety factor, μ is the friction coefficient and r is the radius of the shaft (Budynas and Nisbett 2011). The friction coefficient for aluminium to carbon fibre epoxy was taken as 0.23 (Schön 2004). The required tightening torque on the clamping bolt was calculated with

$$T_{Bolt} = K \times D \times F_c \quad (3)$$

where T_{Bolt} is the tightening torque, K is the Nut Factor of 0.2 for dry, unlubricated steel (Oberg et al. (2020) and D is the nominal bolt diameter. The same approach was used for further clamping points with μ taken as 1.05 for aluminium-to-aluminium contacts. Table 4.1 shows the clamping force and bolt grade requirements (Grampian Fasteners [no date]). M5 bolts were nominally chosen as the default fastener size. Note the safety factor for carbon fibre to aluminium connections ($\mu = 0.23$) were given a safety factor of 1.25 to avoid introducing excessive clamping forces which could crack the carbon fibre shaft since some slight misalignment at the housing was determined to be preferable to catastrophic failure of the main shaft. The weight carrier was given a 5.0 safety factor due to the risk of injury if the weight carrier came loose from the counterweight shaft from an elevated platform and the standard practice of 2.5 was used for the housing clamps. The main shaft assembly was set up as show in figure 4.10 with 2 direct forces on the bolt recesses of the front handle boss and a remote force was positioned at the scanner's location. The scanner load was calculated as the combined weight of all components distal of the locking joint as a worst-case loading scenario. The shaft to boss contact was set as bonded to simulate a rigid clamp. The material selected to simulate the carbon fibre shaft was the epoxy carbon fibre unidirectional prepreg with 230 GPa yield strength fibres from the standard Ansys composite materials library. The material properties are shown in appendix B.

Table 4.1: Clamping force and bolt grade requirement calculation results

Contact	μ	Shaft Diameter / mm	Safety Factor	F_c / N	Bolt Diameter / mm	T_{Bolt} / Nm	Minimum Grade
Carbon fibre-Aluminium	0.23	30	1.25	7609	5	7.6	10.9
Aluminium - Aluminium	1.05	25.4	5.0	5061	5	5.06	8.8
Aluminium - Aluminium	1.05	22.2	2.5	686	5	0.69	4.6



Figure 4.10: v0.3 housing shaft assembly setup.

A maximum deflection of 11.8 mm was observed in the shaft (see appendix B). The maximum stress of 555 MPa occurred in the clamping bolt recess. This was likely due to the low thickness of material in the recess. The obvious remedy of increasing the local material thickness was considered, however, when accounting for the current designs overall weight of 18 kg, adding more material appeared to be a step backwards in design. In addition, the challenges of reworking carbon fibre presented an issue with cutting slots into the main shaft (Liu et al. 2019), so it was more advantageous to cut into the aluminium handle shaft to route the Bowden cable. These two factors inspired a redesign of the main and handle shafts to fit them together concentrically using a cylindrical nylon 6 sleeve to adapt to the two shaft diameters and allow them to fit together rigidly (figure 4.12).

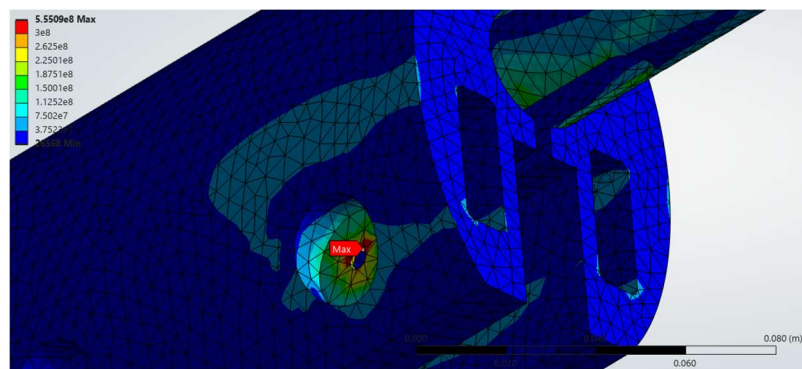


Figure 4.11: v0.3 main shaft assembly maximum stress.

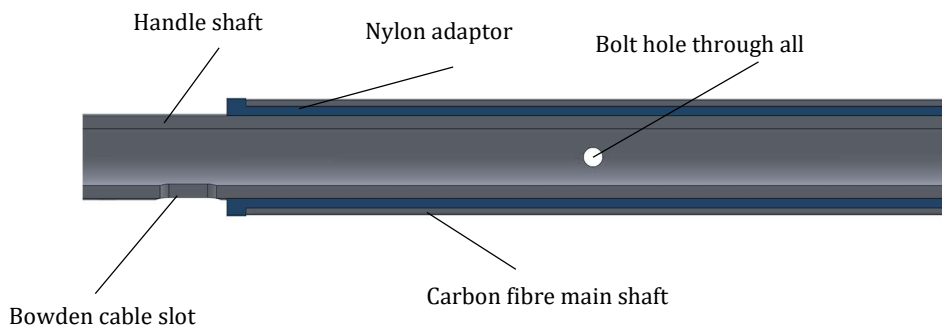


Figure 4.12: Concentric shaft assembly iteration.

This shaft redesign was considered v0.4. The setup and initial stress result for the v0.4 shaft design can be seen in appendix B along with the updated mass, moment and load calculations for the final v0.6 (these values were approximately the same as v0.4, so only v0.6 were included). The x-direction Young's Modulus and x-direction tensile strength were modified in line with the supplier's listed values (Easy Composites Ltd 2024) to 64 GPa and 650 MPa respectively. This modification was aimed to give a more accurate simulation of the material properties compared to the standard Ansys properties. The orthotropic properties in the y and z directions were left unchanged. This would likely result in conservative approximations of the materials shear and interlaminar strength due to the actual shafts' 0, 90 prepreg layup. Appendix B contains the material data edits for the carbon fibre shaft. Figure 4.13 shows the maximum stress concentration of 197 MPa after a local mesh refinement which did not cause any structural concerns. The deflection result in figure 4.14 indicated a much higher maximum deflection relative to v0.3 of 36.6 mm.

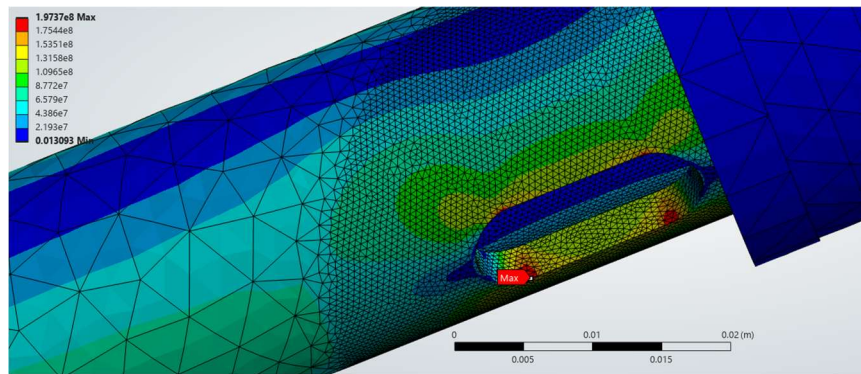


Figure 4.13: Locally refined mesh result at Bowden cable slot for v0.4 shaft assembly.

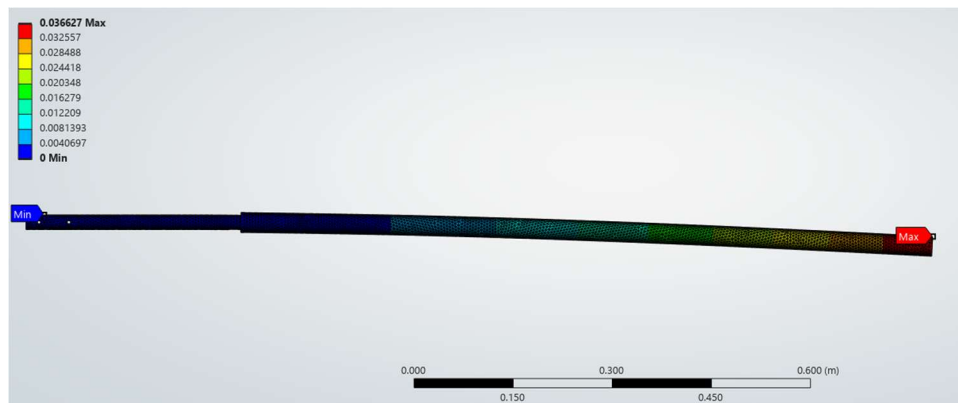


Figure 4.14: Deformation result for v0.4 shaft assembly

The deflection at the distal end of the aluminium handle shaft was approximately 4 mm. If the carbon fibre shaft were assumed to have zero deflection, it would follow a perfect straight line from the end of the aluminium shaft. Pythagoras theorem was used to calculate a theoretical deflection at the end of the carbon fibre shaft of 16 mm under this assumption. Indicating that 20.6 mm of the maximum deflection was a result of the carbon fibre shaft bending. This was still

a large increase from v0.3, likely due to the updated material properties with lower stiffness and strength values. Without the physical shaft available, it was difficult to discern whether this deflection would be a hinderance in the use of the extension, although the stress results validated it was highly unlikely to fail. As a result, another alternative version, v0.5, was made as a potential redesign combining the handle and main shafts into one aluminium shaft at 500 mm length to minimise deflection and reduce the total balanced weight from 18 kg to 12.7 kg. The v0.5 full assembly is shown in appendix A. Other iterative improvements were made in v0.5 which were implemented into the final v0.6 design.

4.1.2 Joint Adaptor and Housing Analysis and Iterative Development

Figure 4.15 shows the v0.3 housing assembly setup to test its ability resist the clamping forces and the weight of the scanner. The scanner weight was applied to a small section at the distal end of the housing as a worst-case loading scenario. The stress results reported no issues except for some excessively thin material at the bolt recess on the housing exterior leading to a large stress concentration. The deformation result was minimal and can be viewed in appendix B.

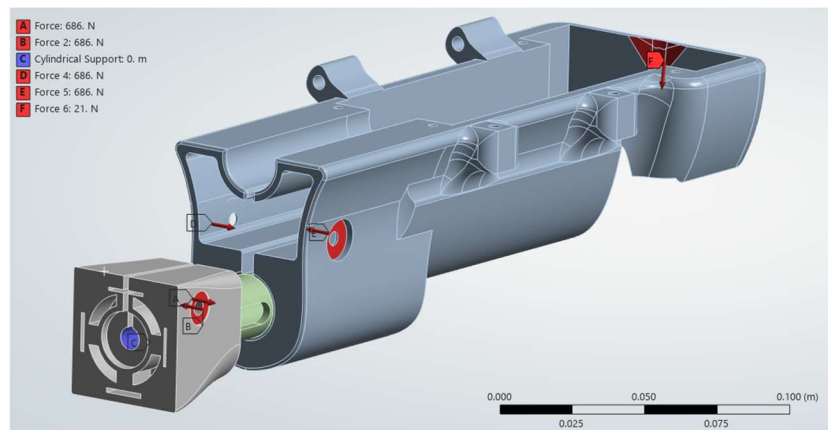


Figure 4.15: Setup for v0.3 housing assembly.

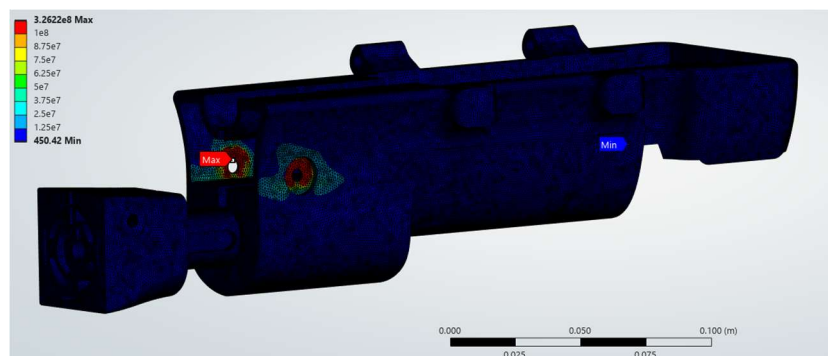


Figure 4.16: Stress result for v0.3 housing assembly.

The bolt recess was thickened in v0.5 and the length of the housing was reduced in line with the concept feedback from MES. Figure 4.17 shows the new housing model and loading setup to test the strength of the housing hatch and the iterative improvement to the housing.

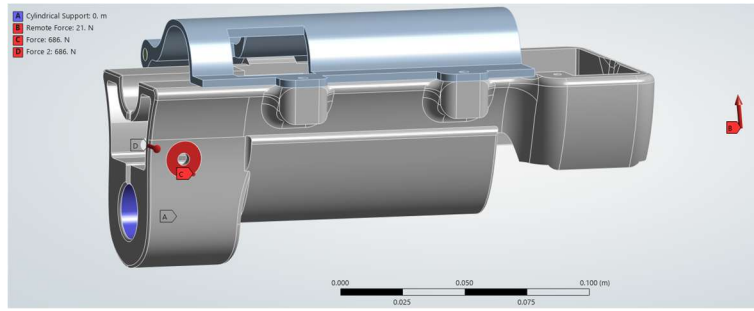


Figure 4.17: v0.5 load setup for housing with hatch

The housing hatch was secured to the housing using a 5 mm steel rod through the hinge. The inner faces of the bolt holes through the hatch and housing were set as bonded contacts to simulate an adequately strong fastener securing the hatch. The scanner weight was applied using a remote force to a small section of the rear of the hatch (figure 4.18) to simulate the end of the moment of the grip pushing upwards on the hatch. Figure 4.19 shows the stress distribution on the hinge rod and figure 4.20 shows the stress over the assembly. The maximum stress of 84 MPa occurred at a sharp corner of a cutout in the hatch, this could be mitigated by adding a fillet to the cutout, however the stress magnitude was well within acceptable limits, hence this housing design was implemented in v0.6.

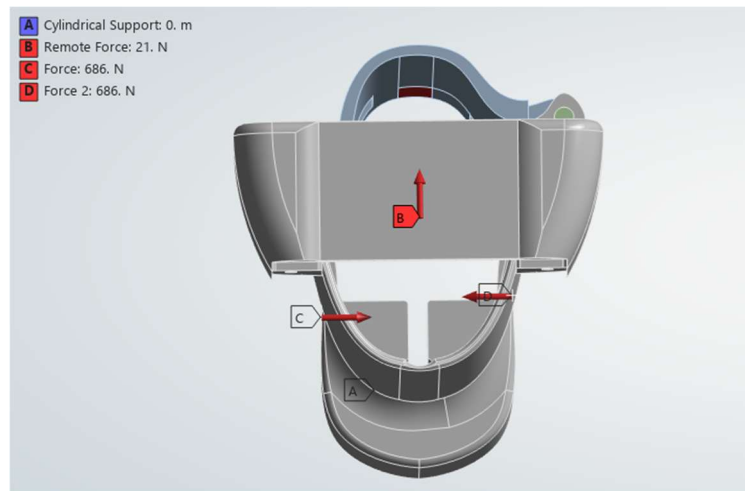


Figure 4.18: v0.5 load setup for housing with hatch

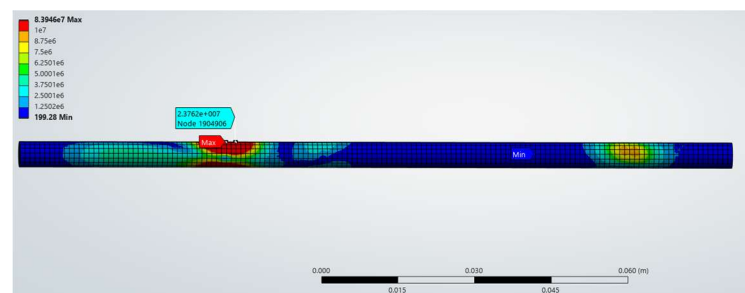


Figure 4.19: v0.5 load setup for housing with hatch

J: v0.5 Housing hatch assembly
Figure
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Unit: Pa
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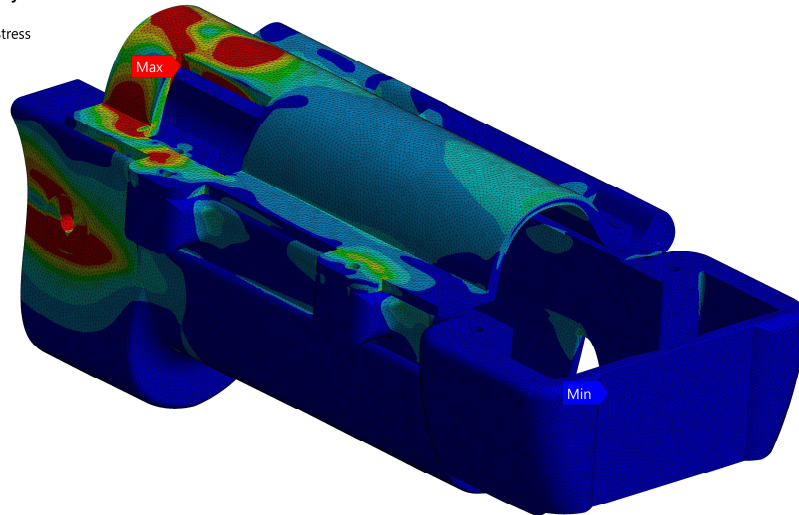
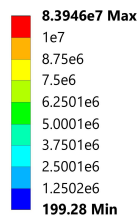


Figure 4.20: Stress result for v0.5 housing with hatch

The v0.3 adaptor for the main shaft to the locking joint was analysed as a single part. Figure 4.21 shows the loading setup where a remote force of all the distal components' weight was applied at the scanner location to only the lower half of the central bolt hole. An indeterminate proportion of the load would be taken up by the adaptor face and potentially the tab slots, so this presented a worst-case scenario in line with the approaches to other components. Figure 4.22 shows the stress result with a maximum stress of 390 MPa at the main shaft insert. A 1mm radius fillet was added to this location, however a new simulation was not set up to test the effect of this change since the locking joint had not yet arrived, therefore it was likely that the adaptor would have to be modified again to accommodate the actual part.

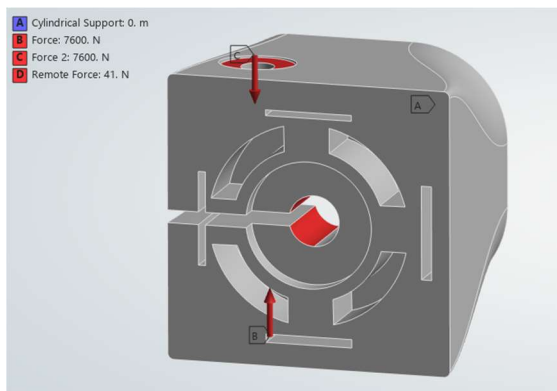


Figure 4.21: v0.3 main shaft adaptor FEA setup

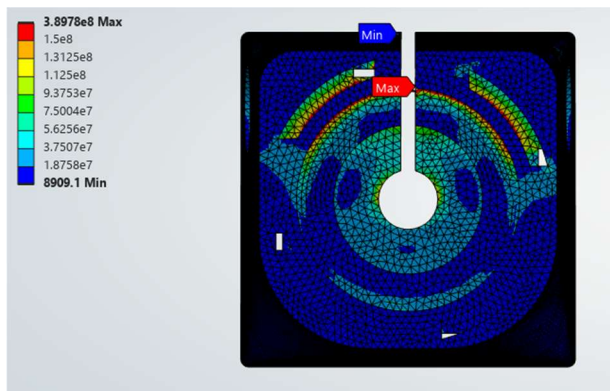


Figure 4.22: v0.3 main shaft adaptor stress result

Figure 4.23 shows the final v0.6 main shaft adaptor design. The length was increased to accommodate the addition of a slot to route the Bowden cable to avoid cutting a slot into the shaft. The tab slots were adjusted to fit the locking joint tabs, and the thickness of the joint mating face

had to be reduced to 5 mm to allow joint's included attachment bolt to reach full thread engagement with its T-slot nut when assembled.

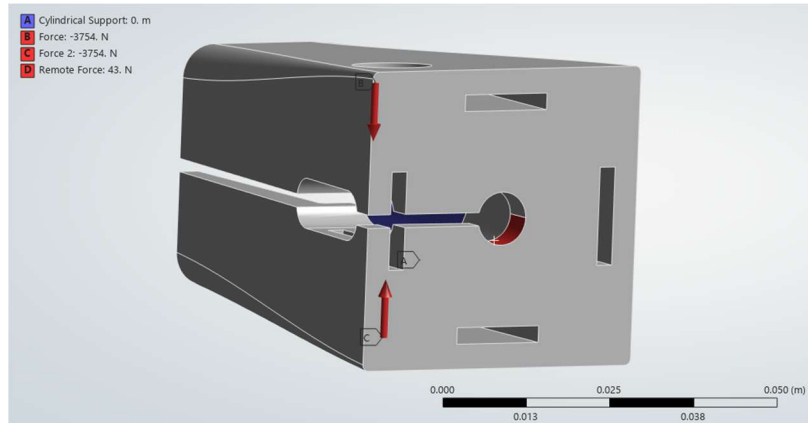


Figure 4.23: v0.6 main shaft adaptor FEA setup.

The results yielded a maximum stress of 721 MPa at the joint bolt hole (figure 4.24). Although this stress was over double the yield strength of a typical aluminium alloy, it only occurred at the surface around the bolt hole and reduced sharply in the surrounding material, indicating local deformation was more likely to occur than structural failure. Figure 4.25 shows the high stresses of approximately 450 MPa also observed at the main shaft fillet. It was determined that the chosen approach with only loading the bolt hole may have been unrealistic and further investigation using an assembly with a more accurately modelled locking joint and attachment bolt should be carried out, in addition to investigating whether the fillet stress was caused by the clamping connection or the attachment bolt load by applying them individually. Due to time constraints the FEA had to be concluded, and this iteration of the adaptor was used in the 3D-printed prototype. Further results for the adaptor are shown in appendix B.

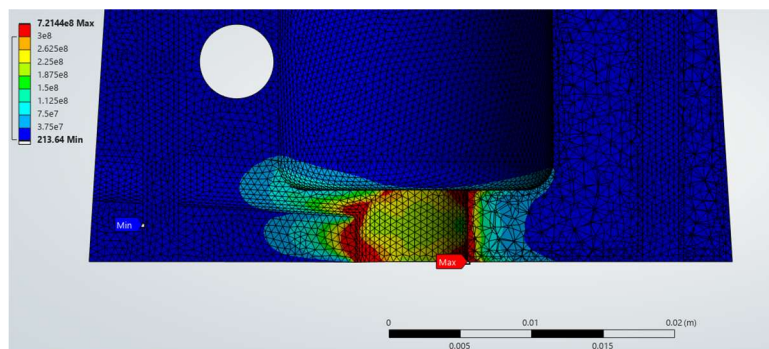


Figure 4.24: v0.6 main shaft adaptor attachment bolt stress concentration shown in a cross section through the split.

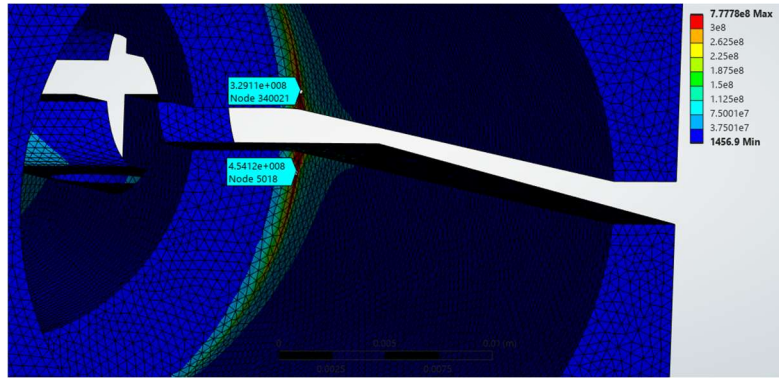


Figure 4.25: v0.6 main shaft adaptor stress concentrations at 1 mm fillet.

4.1.3 Counterweight Analysis

To validate the counterweight design, an assembly of the v0.3 counterweight shaft and weight carrier was set up as shown in figure 4.26. The cylindrical supports mirrored those of the handle shaft in section 4.1.1. The maximum stress of 60 MPa occurred on the shaft along the line of the split in the weight carrier clamp (figure 4.27) and maximum deflection of 3 mm at the weight carrier were recorded. As a result, the v0.3 counterweight design was carried forward to each successive design without alterations except for variation in the shaft length to make the extension less cumbersome to manoeuvre with objects behind the user and the addition of a hip brace (see section 5.1.1). The deflection result and the full assembly stress result are shown in appendix B.

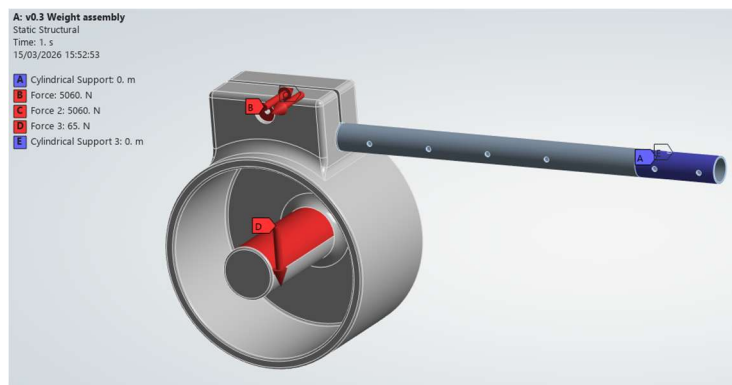


Figure 4.26: v0.3 counterweight assembly setup.

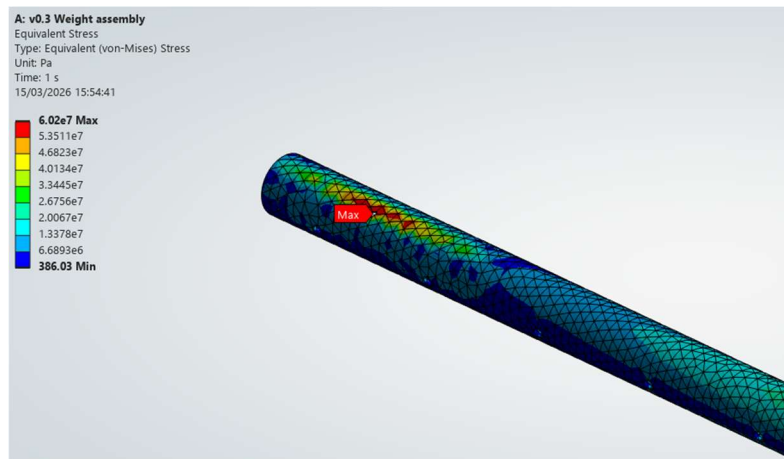


Figure 4.27: v0.3 counterweight assembly maximum stress.

4.2 Mechanical Testing of the Carbon Fibre Shaft

Once the carbon fibre shaft was delivered, it was tested in a loaded cantilever test to measure and compare the actual deflection of the main shaft in FEA to the actual shaft. The testing setup is shown in figure 4.28. Two lengths of scrap aluminium tubing were used to support and load the carbon fibre shaft. The aluminium support shaft inserted loosely into the carbon fibre shaft and clamped in place with a vice. Masses were attached to the load shaft at a consistent 300 mm from the end of the carbon fibre and the distance from the top face of the countertop to the lower side of the end of the carbon fibre shaft was measured at using a ruler. The outer diameters of the aluminium shafts were measured at approximately 25 mm using vernier callipers and the rated internal diameter of the carbon fibre shaft was 27 mm, so the shafts were a loose fit. The mass of the weight carrier was 2.22 N and a total of 50 N of masses were used to for loading. The initial countertop distance was measured with only the weight carrier attached as a baseline reading. To give a better comparison to the FEA due to the difference in the load offset distance from the shaft end, the moment of each load about the end of the carbon fibre shaft was calculated. Figure 4.29 shows a plot of deflection against end moment. The full data is available in appendix B.

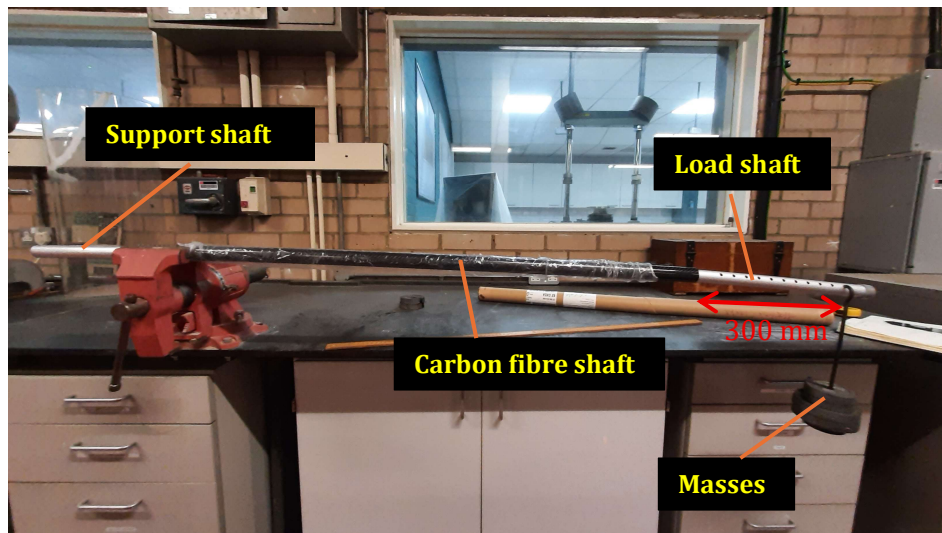


Figure 4.28: Carbon fibre shaft mechanical testing apparatus.

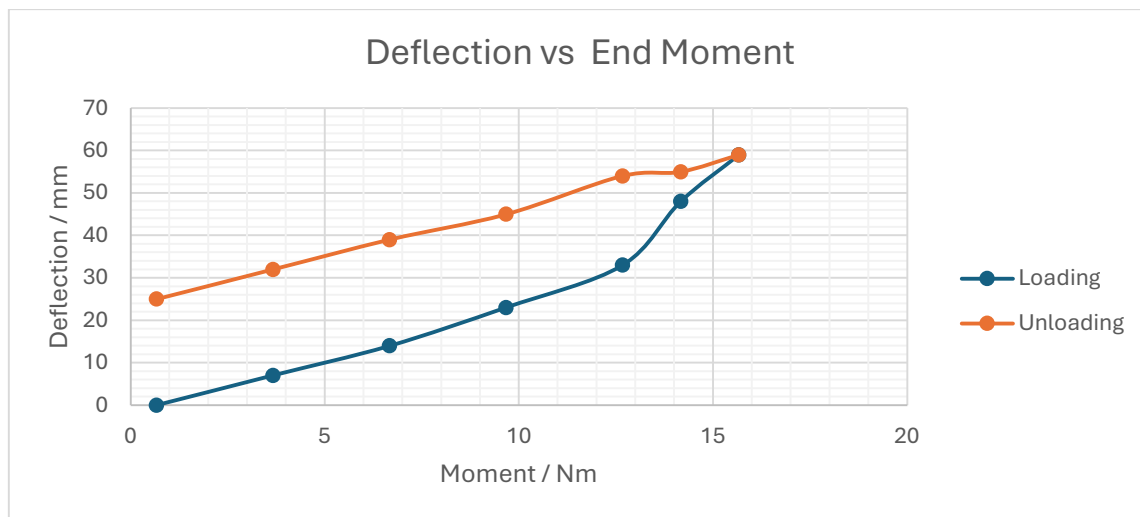


Figure 4.29: Deflection against moment for full loading and unloading data.

The loading phase showed an initially linear gradient until a moment of approximately 12.5 Nm, where the gradient increased and becomes non-linear. In the unloading phase the reverse occurred with an initial non-linear section followed by a return to a linear gradient. Once unloaded fully with the weight carrier still attached, a final deflection of 25 mm was recorded. Since there was no permanent deformation from bending or damage around the circumference of the ends of the carbon fibre, this final deflection was likely due to one or both aluminium shafts shifting or tilting during loading due to the clearance between them. The final deflection was therefore assumed to be a systematic error in the unloading data. The linear sections of the loading and unloading data were plotted separately with trendlines and their respective equations calculated using excel (figure 4.30). Taking the v0.4 loading setup in FEA as a 14.6 Nm moment (43 N at 0.34 m from the shaft end) and inputting it into the loading and unloading equations predicted 37.1 mm and 57.8 mm respectively. After calibrating the unloading result by subtracting the 25 mm error, the result was 32.8 mm. The average of the loading and calibrated

unloading deflections was 35.0 mm, 1.6 mm or 4.6 % lower than the predicted deflection from FEA. Due to the rigidity of the shaft and the exaggerated loading methodology of combining the weight of all the distal components into one point, it was concluded that the carbon fibre shaft would be adequate for the final design and was preferable to the v0.5 aluminium shaft since the carbon fibre provided roughly twice the reach at a similar overall extension weight.

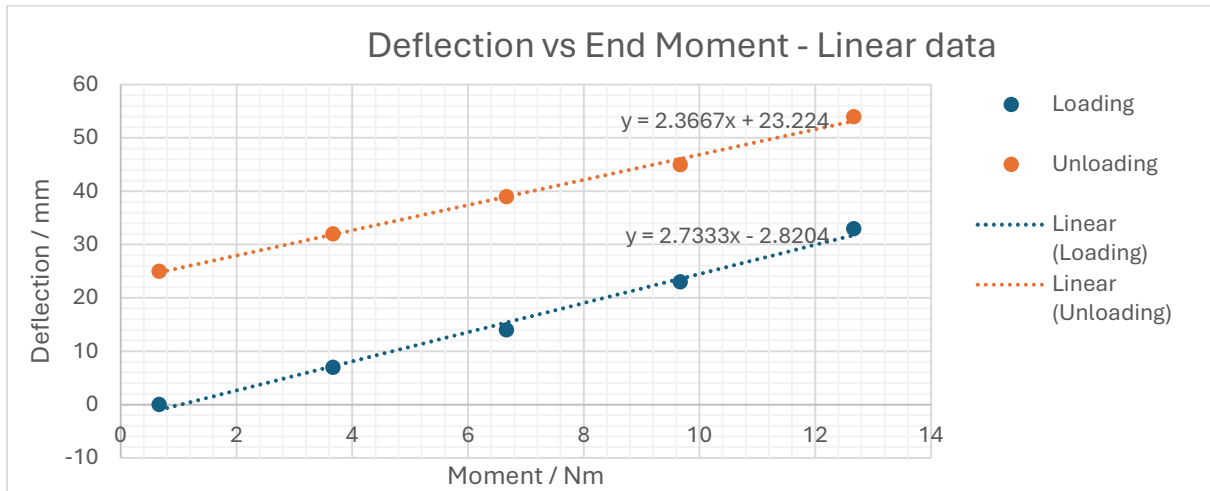


Figure 4.30: Trendlines and respective equations for the linear loading and unloading data.

5. FINAL DESIGN AND CRITICAL EVALUATION

5.1 Final Design

5.1.1 Final Design Alterations

Figure 5.1 shows a 3D render of the final v0.6 design at the time of writing. This section details the final additions to the design after validation and testing were concluded.

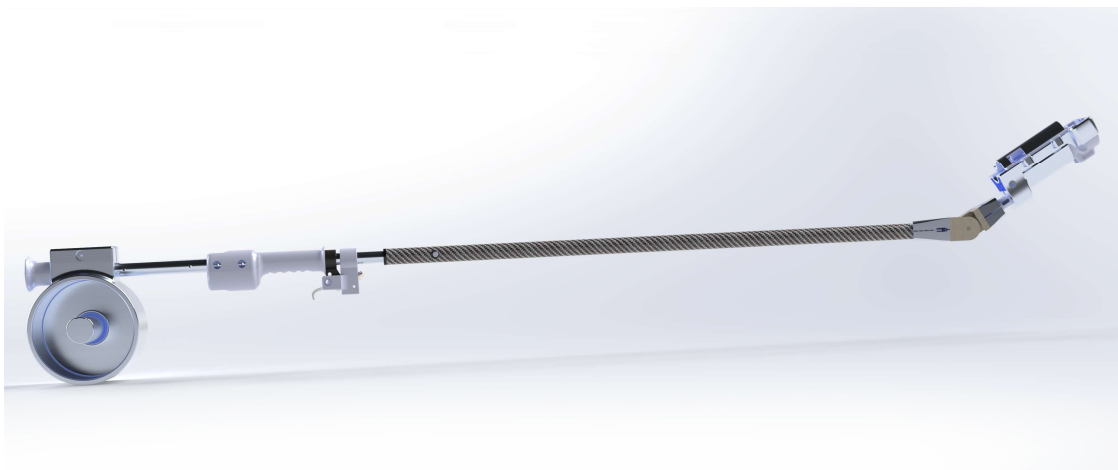


Figure 5.1: 3D render of the final v0.6 design.

Firstly, since the final shaft design required a bolt through the shafts, a protective insert was modelled to mitigate the potential edge wear at the bolt hole in the carbon fibre shaft (figure 5.2).

Figure 5.3 shows the design for an ergonomic trigger to replace the default clutch lever. This addition was made since the lever would require the user to actuate it with the last three digits, while holding the grip between the index finger and thumb. The actuation the new lever could be achieved with only the index finger (figure 5.4) allowing for a more secure grip between the last three fingers and thumb.



Figure 5.2: Protective inserts for carbon fibre shaft.

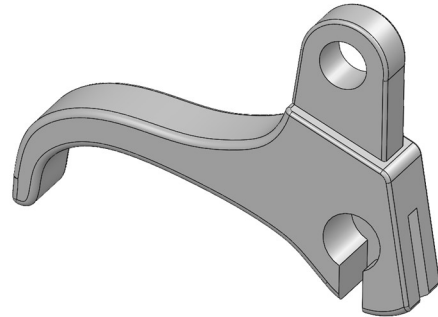


Figure 5.3: Custom trigger style clutch lever.

Since the front handle boss was no longer structural as of v0.4, its size was reduced to a simple attachment and the trigger locking mechanism, which had seen limited development since v0.2, was redesigned to accommodate the procured spring plunger (figure 5.5). It featured slots for the plunger's included nut to slot into on either side before being vertically constrained when the plunger was threaded through the nut. The spring plunger was planned to be manually adjusted so it would catch into a hole drilled into the side face of the trigger which aligned with the plunger tip at full actuation of the trigger. At the time of writing, this work was not completed due to complications with the assembly process. Figure 5.6 shows the plunger installed in the 3D-printed attachment.

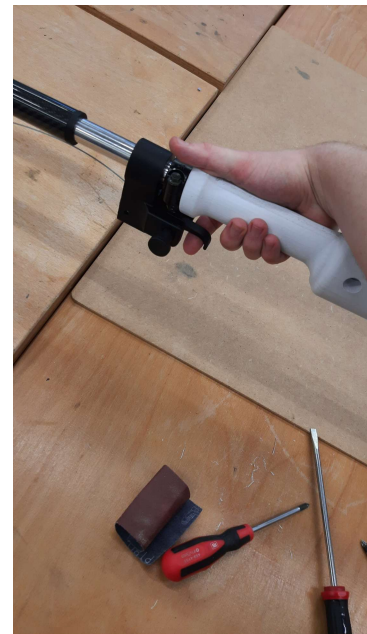


Figure 5.4: Custom trigger actuation.

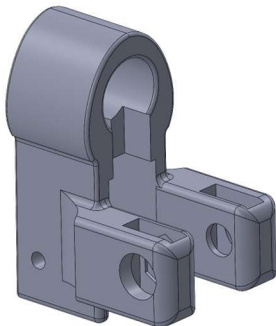


Figure 5.5: New front handle attachment.



Figure 5.6: Front handle attachment 3D print and spring plunger

Figure 5.7 shows the hip brace added to v0.5 and carried forward into the final design. The brace was pressed onto the end of the counterweight shaft and was intended to provide a comfortable rest for the extension if the user held it in front of their body to maximise reach. Figure 5.8 shows a highlighted late stage modification to the housing of an additional flange to attach the threaded adjuster for the bowden cable since there was not enough clearance in the trigger enclosure of the housing interior to attach it.



Figure 5.7: Hip brace.

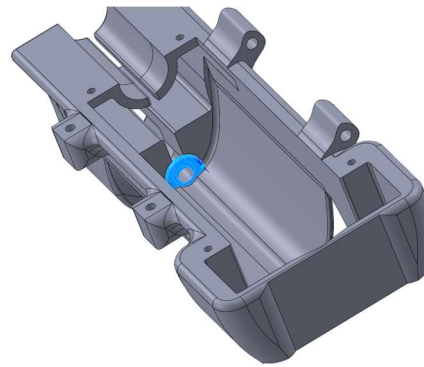


Figure 5.8: Flange added to housing exterior.

The final addition was a cable management ring shown in figure 5.9 to be 3D-printed to tightly fit over the main shaft and remove the need to cut several holes along the main shaft and potentially compromise its structure.



Figure 5.9: Cable management ring

5.1.2 Prototype Assembly

Figure 5.10 shows the assembled prototype in its current state at the time of writing.



Figure 5.10: Current-stage prototype assembly.

The required holes and slots from the CAD model were machined into the shafts and the weight carrier, handle grip, custom trigger, front handle attachment, protective inserts, adaptors, housing exterior and housing interior were 3D-printed in PLA. A replica AP21 handle shown in figure 5.11 was also printed in PLA to test the fitment of the housing.

Figures 5.12 through 5.14 show the installation of the Bowden cable through the prototype. The supplier's website advertised the cable kit as having solderless nipple attachment, however, there was no apparent solderless attachment method included in the kit; hence the nipples were soldered onto the cables. The handle front attachment was not installed while the cable was being run and soldered. Since the nipple was too large to fit through the slot in the aluminium shaft, the shafts could not be axially pulled apart without de-soldering or cutting off the nipple and potentially having to use the last available length of cable. As a result, the work on attachment and locking mechanism was not completed in time for the project end.



Figure 5.11: Replica AP21 grip.



Figure 5.12: Prototype housing



Figure 5.13: Prototype handle shaft cable slot and soldered nipple.

Both of the 3D-printed joint adaptors cracked as the clamping bolts were tightened (figure 5.14). This resulted in the prototype being very structurally weak at the distal end. It was clearly not strong enough to test with weights attached to the replica AP21 grip, therefore it could not be validated for live testing with the AS1 scanner without a redesign of the adaptor attachment method.

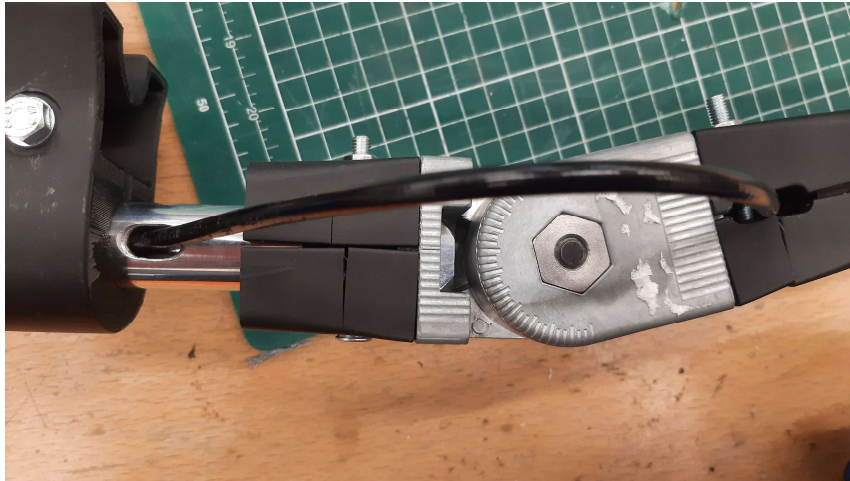


Figure 5.14: 3D-printed prototype joint adaptors.

5.2 Functionality Review

The compression spring ordered from RS components was too stiff and made the handle trigger excessively heavy. A replacement spring was sourced from the university workshop with similar dimensions and a much lower stiffness. This allowed for the full range of motion of the trigger actuator (figure 5.15) to be achieved through the handle trigger. However, some noticeable resistance could still be felt through the trigger.



Figure 5.15: Trigger actuator installed in the housing.

The prototype was briefly tested with masses inside the weight carrier as shown in figure 5.16 (masking tape was used to keep them in place since standard 2" weights were not available). The extension was already noticeably front heavy without masses added. The shoulder strap (figure 5.17) was able to partially mitigate this issue, making the extension easier to carry. With 2 kg of

mass added, the extension was much better balanced and with 3 kg it was noticeably back heavy. The total mass of the prototype was measured at 2.95 kg. Comparing to the final designs approximate 11.8 kg mass (without the scanner) if aluminium components were used represents a 300 % increase.



Figure 5.16: *Weight carrier testing setup.*



Figure 5.17: *Shoulder strap attachment*

5.3 Future Work

Several aspects of the extension design could be improved and optimised further. The most immediate improvement would be to complete the prototype locking mechanism installation by first de-soldering the trigger side cable nipple. Further improvements to the locking mechanism would involve remodelling the custom trigger to include a ramped slot into its side leading to a recess which would allow the spring plunger to be depressed gradually during a trigger pull before springing back into the recess to lock the trigger in position. The fine tuning of the locking mechanism and general ease of assembly would be improved by using solderless cable nipples. In addition, it could allow the cable to have a more optimal length with less bending since it could be adjusted at will. This could result in a smoother trigger pull and reset since a higher bending angle will decrease the mechanical efficiency of a Bowden cable (Hofmann et al. 2018).

Reconsidering the design of the attachment method for the joint adapters and housing in favour of a combined approach of split clamps and bolting directly through the shafts could yield several benefits. The split clamp would still reduce the reliance on tight tolerances for a rigid connection; however, the bolts would remove the requirement for large clamping forces to tangentially constrain the shafts. This could allow the weight of the aluminium components to be reduced, or alternatively, allow the consideration of a different material with a lower yield strength. Either option could allow a reduction of the total weight of the extension, bringing the design closer to the ergonomics of the current prototype. The mass of the aluminium weight carrier should also

be reduced and substituted for an increase in the number of counterweight masses since the cost of the masses would be substantially less per unit mass than that of a custom aluminium alloy component. Attaching a support strap between the counterweight shaft and a belt or harness to further distribute the load over the user's body could be considered a potential mitigation if the total weight of the extension cannot be reduced, or another ergonomic enhancement in addition to the shoulder strap. To complete the design, more consideration should be given to fastener selection to avoid excess bolt lengths from becoming a hazard to the user. Fastener diameter should also be reconsidered as some of the lengths required may be less common at the nominally selected M5 size. This could also allow the same grade of fastener to be used throughout the assembly.

6 CONCLUSIONS

This project successfully translated a design concept into a fully modelled extension handle through six iterations featuring a non-invasive attachment method to the existing device, a functional trigger linkage system and a counterweight to balance the extension. Finite element analysis was carried out to validate the structural integrity of the shafts. The analysis produced a deformation result within a 4.6% margin of the physical testing result of the carbon fibre shaft.

The physical prototype proved that a Bowden cable operated trigger linkage system was viable and the theoretical ergonomic requirements were successfully translated into physical design features such as the addition of the hip brace, shoulder strap and custom index finger trigger, all of which led to a noticeable improvement in the economics of the prototype. The physical assembly demonstrated the effectiveness of the counterweight system since a small mass of approximately 2 kg could balance the 2.95 kg prototype. Some drawbacks with the design were revealed in the assembly process, including the soldering requirement of the Bowden cable which prevented easy disassembly and the friction-based split clamps required excessive clamping force to function which resulted in the 3D-printed joint adapters cracking.

Future prototyping should incorporate solderless cable nipples to allow Easier disassembly and fine tuning of the trigger linkage to optimise the bending angles and resulting mechanical efficiency. In addition, the custom trigger should be remodelled to include a ramped slot to improve the locking mechanism design.

The joint adaptors should adopt a hybrid split-clamp and through-bolt attachment method design to reduce the required clamping forces and allow weight reduction through reduced part volume or selection of a lower density, lower strength material. The aluminium weight carrier should also be reduced in volume to reduce the manufacturing cost, relying on mass of additional weights instead of the mass of the carrier as a counterweight. Additionally, the fastener sizes should be

reconsidered and standardised to prevent excess thread lengths becoming a hazard. If the total mass of the design after further iteration cannot be reduced, a waist belt or harness in addition to the shoulder strap should be considered for better load distribution.

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APPENDICES

Appendix A: Additional Design Methodology Materials

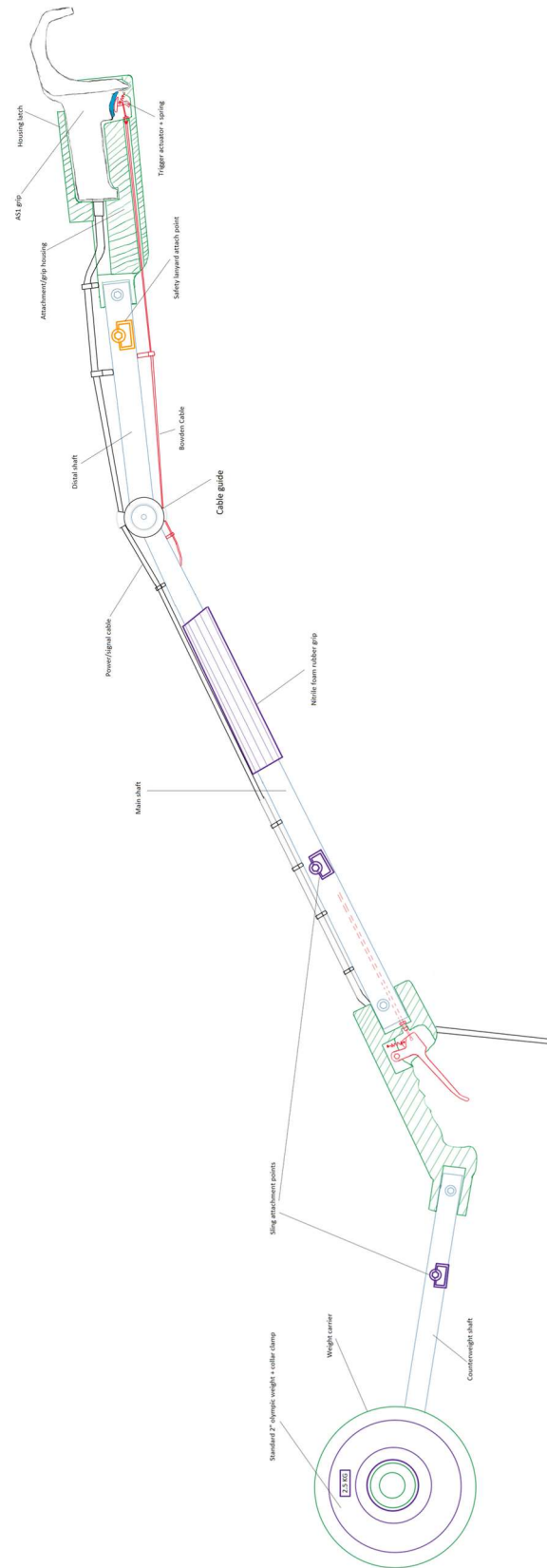


Figure A1: Full 2D concept presented to MES

Table A1: v0.5 moment calculations from CAD assembly

v0.5 Parts						
Part	Volume (mm ³)	Material	Density (kg/m ³)	Mass (kg)	Distance (m)	Moment (Nm)
Carrier	1874338.45	Al	2700	5.06071382	0.47	23.3334332
Carrier shaft	37902.22	Al	2700	0.10233599	0.42	0.42164476
Handle Boss	572221.72	PLA	250	0.14305543	-0.03	-0.0421012
Main Shaft	166478.36	Al	2700	0.44949157	0.25	1.10237808
Adaptor 1	48845.38	Al	2700	0.13188253	0.5	0.64688379
Joint (Approx)	131735.26	Diecast Zinc	7600	1.00118798	0.58	5.69655935
Adaptor 2	48845.38	Al	2700	0.13188253	0.62	0.8021359
Housing shaft	16496	Al	2700	0.0445392	0.68	0.2971121
Housing ext	255684	Al	2700	0.6903468	0.84	5.68873377
Housing int	335461.66	PLA	250	0.08386542	0.84	0.69108457
Housing Hatch	49485.91	Al	2700	0.13361196	0.8	1.04858664
Scanner		N/A	N/A	2.14	0.95	19.94373
				Solid PLA: 1250		
				Assumed 20 % infill		

Table A2: v0.6 moment calculations from CAD assembly

v0.6 Parts						
Part	Volume (mm ³)	Material	Density (kg/m ³)	Mass (kg)	Distance (m)	Moment (Nm)
Hip brace	64941	PLA	250	0.0162353	-0.721	-0.1148321
Carrier	1874338.45	Al	2700	5.0607138	-0.615	-30.532046
Carrier shaft	59372.33	Al	2700	0.1603053	-0.584	-0.9183954
Handle grip	884229	PLA	250	0.2210573	-0.318	-0.6896058
Handle shaft	108120	Al	2700	0.291924	-0.141	-0.4037922
Nylon sleeve	46018	Nylon 6	1150	0.0529207	0	0
Main Shaft	N/A	CF	N/A	0.1938	0.41	0.779483
CF joint adaptor	70426	Al	2700	0.1901502	0.949	1.7702394
Joint (Approx)	131735.26	Diecast Zinc	7600	1.001188	1.01	9.9198706
Al joint adaptor	45122	Al	2700	0.1218294	1.06	1.2668552
Housing shaft	16496	Al	2700	0.0445392	1.11	0.4849918
Housing ext	255684	Al	2700	0.6903468	1.27	8.6008237
Housing int	335461.66	PLA	250	0.0838654	1.3	1.0695356
Housing Hatch	45406	Al	2700	0.1225962	1.25	1.5033359
Scanner	N/A	N/A	N/A	2.14	1.35	28.34109
				Solid PLA: 1250		
				Assumed 20 % infill		

Table A3: v0.6 counterweight and total mass calculations

v0.6 Balance - Moments about main shaft bolt						
Total Forward moment	Total Backwards moment	Balance moment require	Counterweight mass / kg			
53.7362252	-32.65867103	21.07755416	3.493623425			
Shaft	Length / m	Selected t / mm	Selected OD / mm	Torque / Nm	Bending Moment / Nm	
Carrier	0.5	1.6	25.4	13.47704751	-31.45044098	
Handle	0.3	3.2	22.2	14.02323899	53.332433	
Main	1	3	30	14.02323899	53.7362252	
Housing	0.08	3.2	22.2	2.979109055	6.931666508	
Sum of mass distal from joint	4.204364991	43N	Total counterweight load		85.64991103	
Joint to scanner dist:	0.34		Counterweight acting dis 0.38 m			
Material properties	Tensile limit / Mpa	Shear limit / Mpa				
CF	650	30				
AL	340	210				
Total mass	13.88509492					

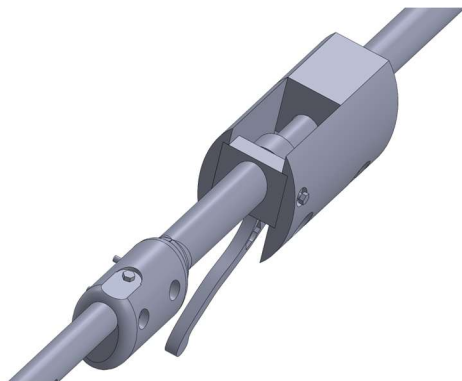


Figure A2: v0.3 handle assembly

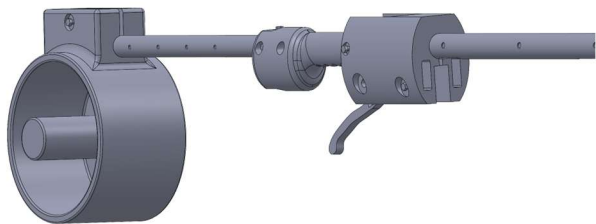


Figure A3: v0.3 handle and weight carrier

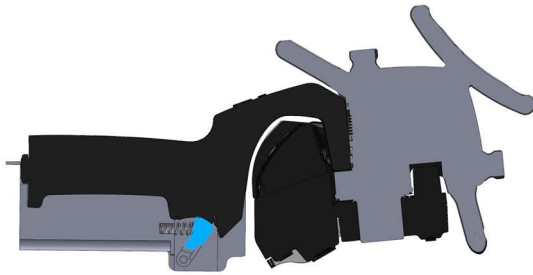


Figure A4: v0.3 housing interior and trigger actuator cross-section.

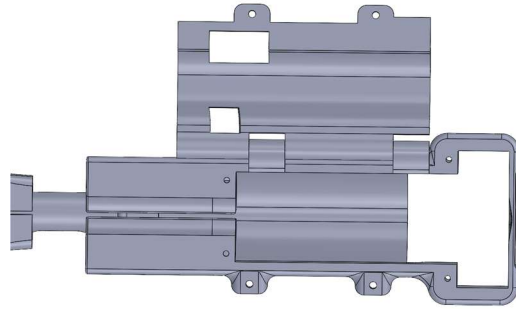


Figure A5: v0.3 housing exterior and securing hatch cross-section.

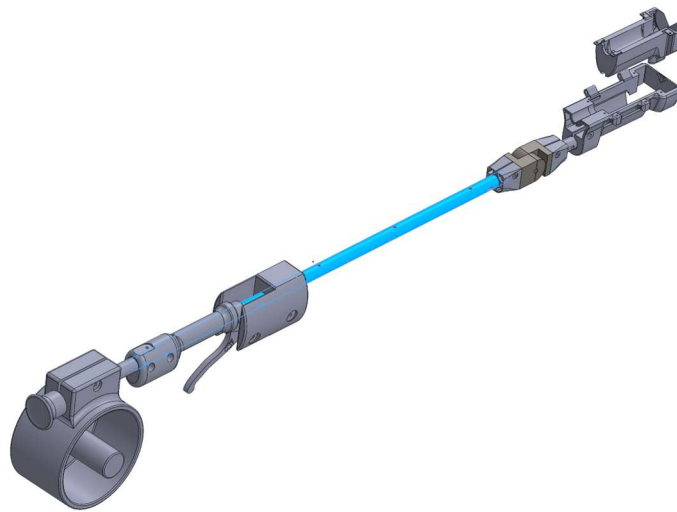


Figure A6: v0.5 full assembly with highlighted handle/main shaft.

Appendix B: Additional FEA Materials

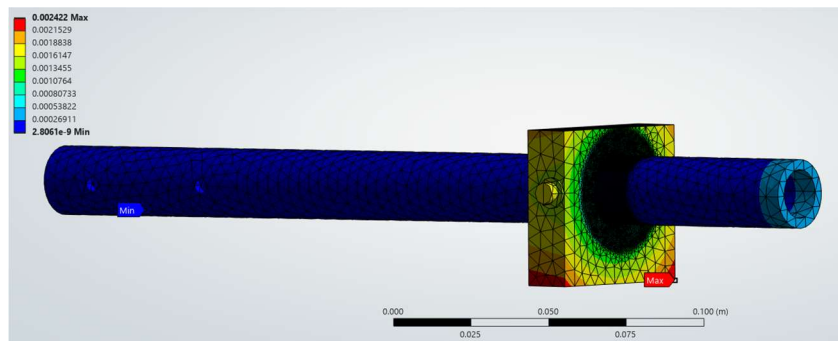


Figure B1: v0.3 handle shaft assembly deformation result

Properties of Outline Row 6: Epoxy Carbon UD (230 GPa) Prepreg 2				
	A	B	C	D E
1	Property	Value	Unit	
2	Density	1490	kg m ⁻³	
3	Orthotropic Secant Coefficient of Thermal Expansion			
8	Orthotropic Elasticity			
9	Young's Modulus X direction	1.21E+11	Pa	
10	Young's Modulus Y direction	8.6E+09	Pa	
11	Young's Modulus Z direction	8.6E+09	Pa	
12	Poisson's Ratio XY	0.27		
13	Poisson's Ratio YZ	0.4		
14	Poisson's Ratio XZ	0.27		
15	Shear Modulus XY	4.7E+09	Pa	
16	Shear Modulus YZ	3.1E+09	Pa	
17	Shear Modulus XZ	4.7E+09	Pa	
18	Orthotropic Stress Limits			
19	Tensile X direction	2.231E+09	Pa	
20	Tensile Y direction	2.9E+07	Pa	
21	Tensile Z direction	2.9E+07	Pa	
22	Compressive X direction	-1.082E+09	Pa	
23	Compressive Y direction	-1E+08	Pa	
24	Compressive Z direction	-1E+08	Pa	
25	Shear XY	6E+07	Pa	
26	Shear YZ	3.2E+07	Pa	
27	Shear XZ	6E+07	Pa	
28	Orthotropic Strain Limits			
38	Tsai-Wu Constants			
42	Puck Constants			

Figure B2: Default material properties of unidirectional carbon fibre shaft in Ansys

Properties of Outline Row 5: Epoxy Carbon UD (230 GPa) Prepreg				
	A	B	C	D E
1	Property	Value	Unit	
2	Density	1490	kg m ⁻³	
3	Orthotropic Secant Coefficient of Thermal Expansion			
8	Orthotropic Elasticity			
9	Young's Modulus X direction	6.4E+10	Pa	
10	Young's Modulus Y direction	8.6E+09	Pa	
11	Young's Modulus Z direction	8.6E+09	Pa	
12	Poisson's Ratio XY	0.27		
13	Poisson's Ratio YZ	0.4		
14	Poisson's Ratio XZ	0.27		
15	Shear Modulus XY	4.7E+09	Pa	
16	Shear Modulus YZ	3.1E+09	Pa	
17	Shear Modulus XZ	4.7E+09	Pa	
18	Orthotropic Stress Limits			
19	Tensile X direction	6.5E+08	Pa	
20	Tensile Y direction	2.9E+07	Pa	
21	Tensile Z direction	2.9E+07	Pa	
22	Compressive X direction	-1.082E+09	Pa	
23	Compressive Y direction	-1E+08	Pa	
24	Compressive Z direction	-1E+08	Pa	
25	Shear XY	6E+07	Pa	
26	Shear YZ	3.2E+07	Pa	
27	Shear XZ	6E+07	Pa	
28	Orthotropic Strain Limits			
38	Tsai-Wu Constants			
42	Puck Constants			

Figure B3: Adjusted material properties of unidirectional carbon fibre shaft in Ansys

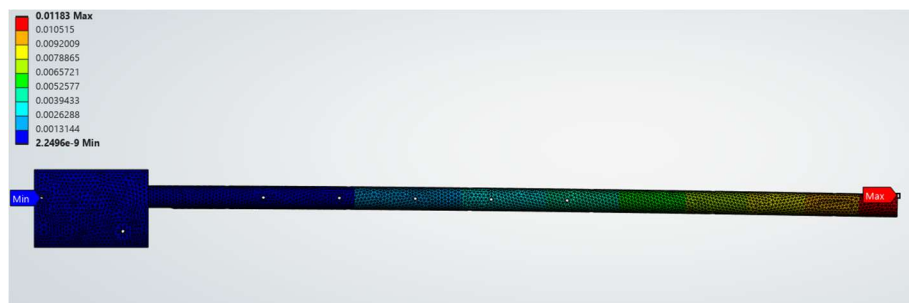


Figure B4: v0.3 shaft assembly deformation result

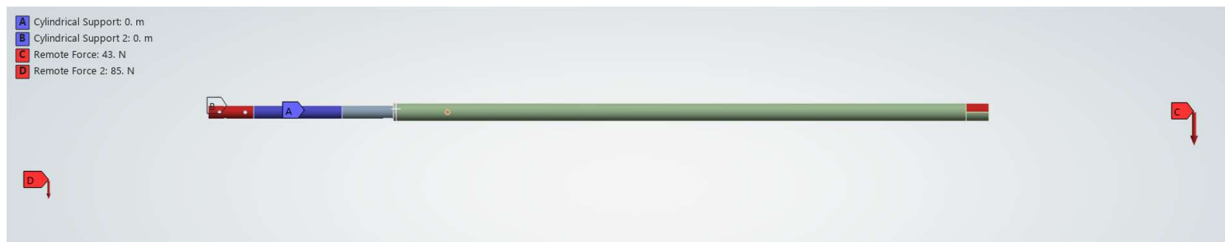


Figure B5: v0.6 concentric shaft redesign FEA setup.

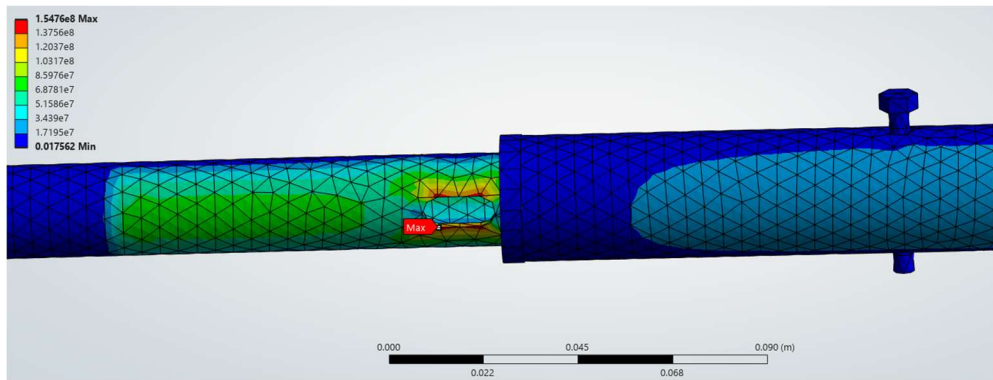


Figure B6: v0.6 concentric shaft redesign pre-refinement result.



Figure B7: v0.3 housing assembly deformation result.

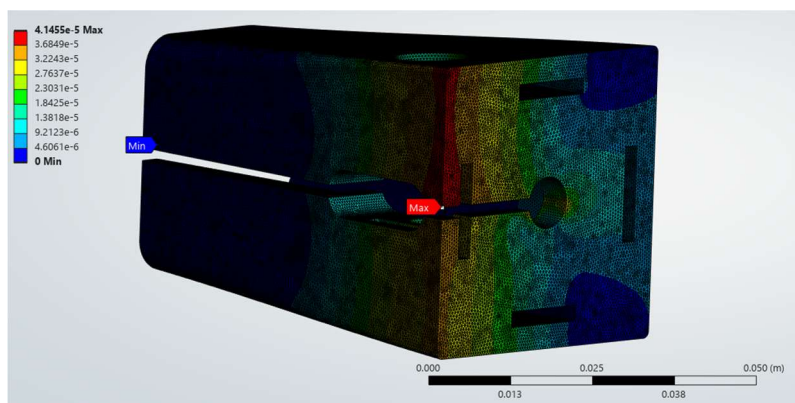


Figure B8: v0.6 main shaft adaptor deformation.

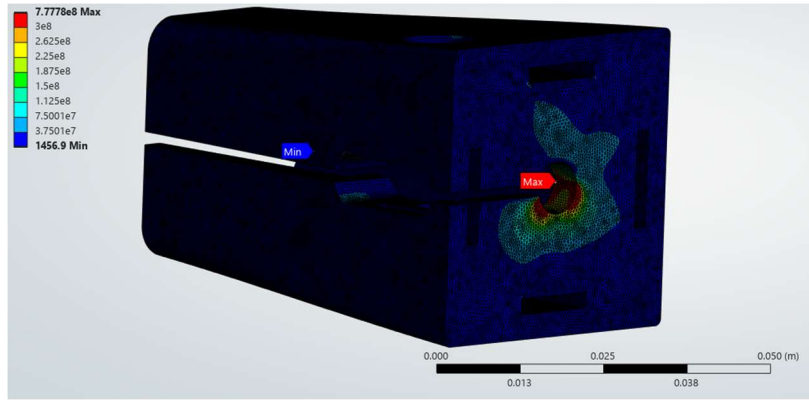


Figure B9: v0.6 main shaft adaptor full stress result.

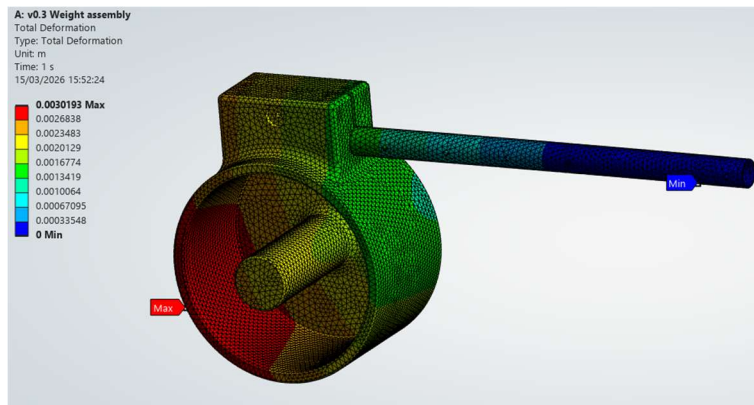


Figure B10: v0.3 counterweight assembly deformation result.

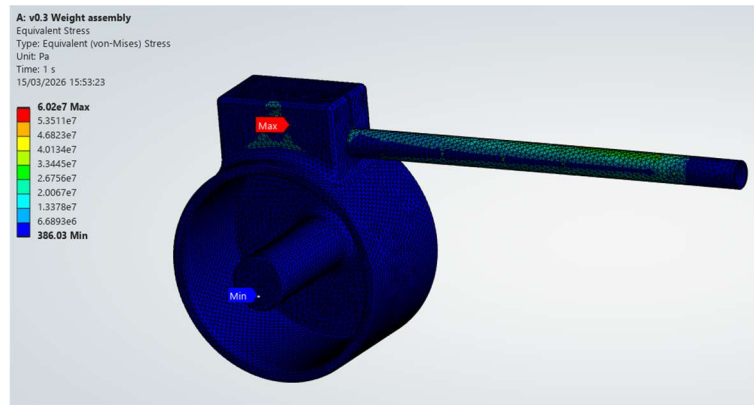


Figure B11: v0.3 counterweight assembly full stress result.

Table B1: Mechanical testing data for the carbon fibre shaft.

	Attached mass weight / N	Total Load / N	Perpendicular distance from shaft end to load / mm	Deflection / mm	Shaft end to countertop vertical distance / mm	Moment / Nm
Loading	0	2.22	300	0	190	0.666
	10	12.22	300	7	183	3.666
	20	22.22	300	14	176	6.666
	30	32.22	300	23	167	9.666
	40	42.22	300	33	157	12.666
	45	47.22	300	48	142	14.166
	50	52.22	300	59	131	15.666
Unloading	45	47.22	300	55	135	14.166
	40	42.22	300	54	136	12.666
	30	32.22	300	45	145	9.666
	20	22.22	300	39	151	6.666
	10	12.22	300	32	158	3.666
	0	2.22	300	25	165	0.666

Appendix C: Final Design Drawings

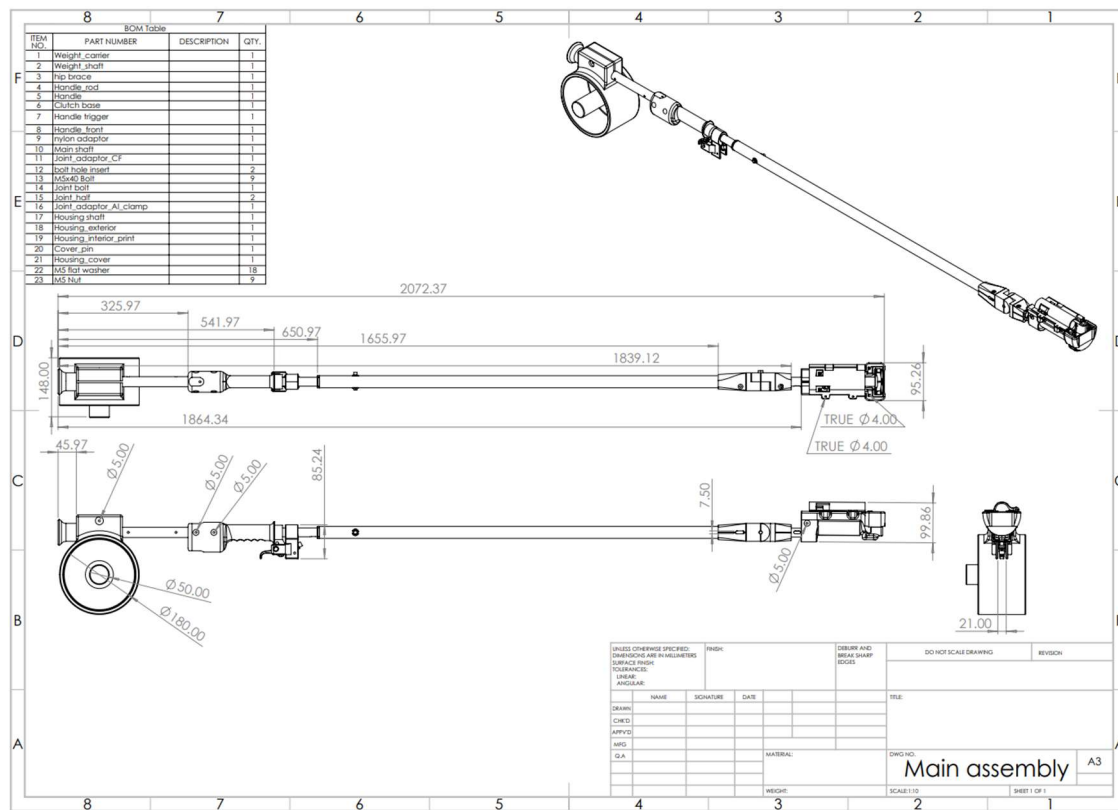


Figure C1: v0.6 main assembly drawing

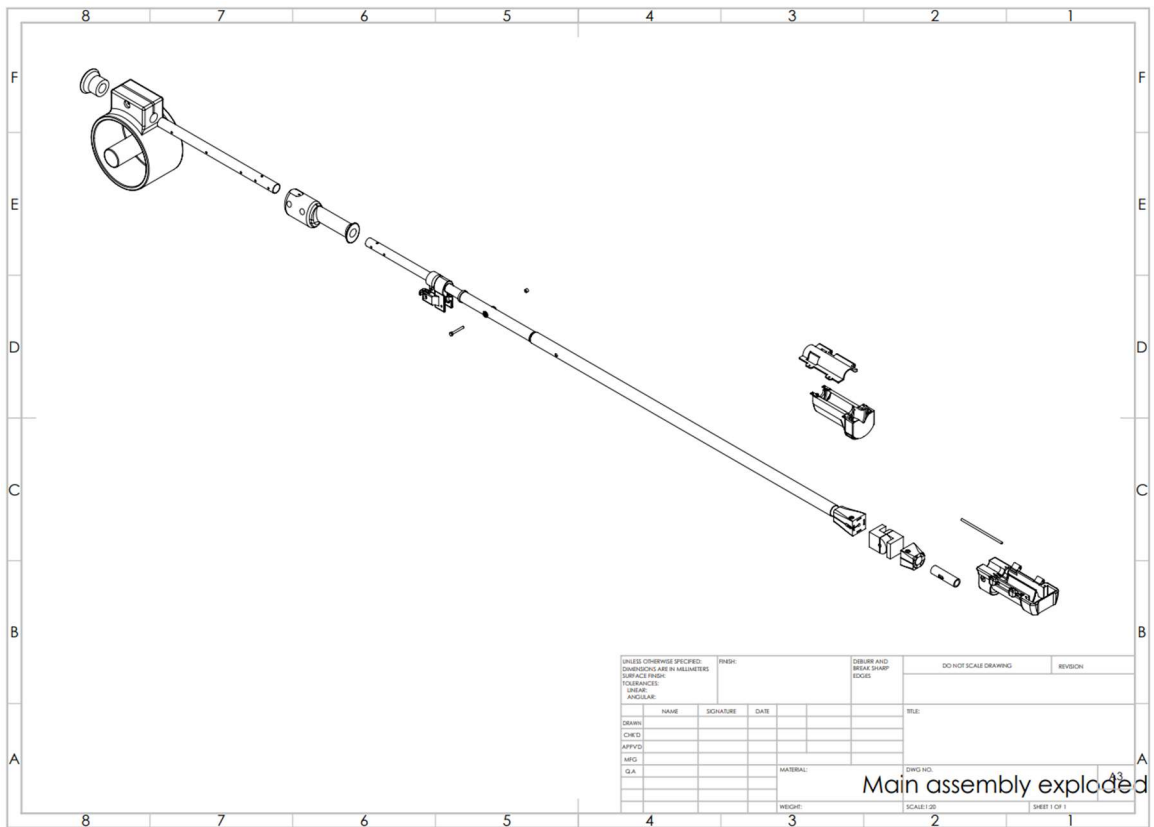


Figure C2: v0.6 main assembly exploded view drawing

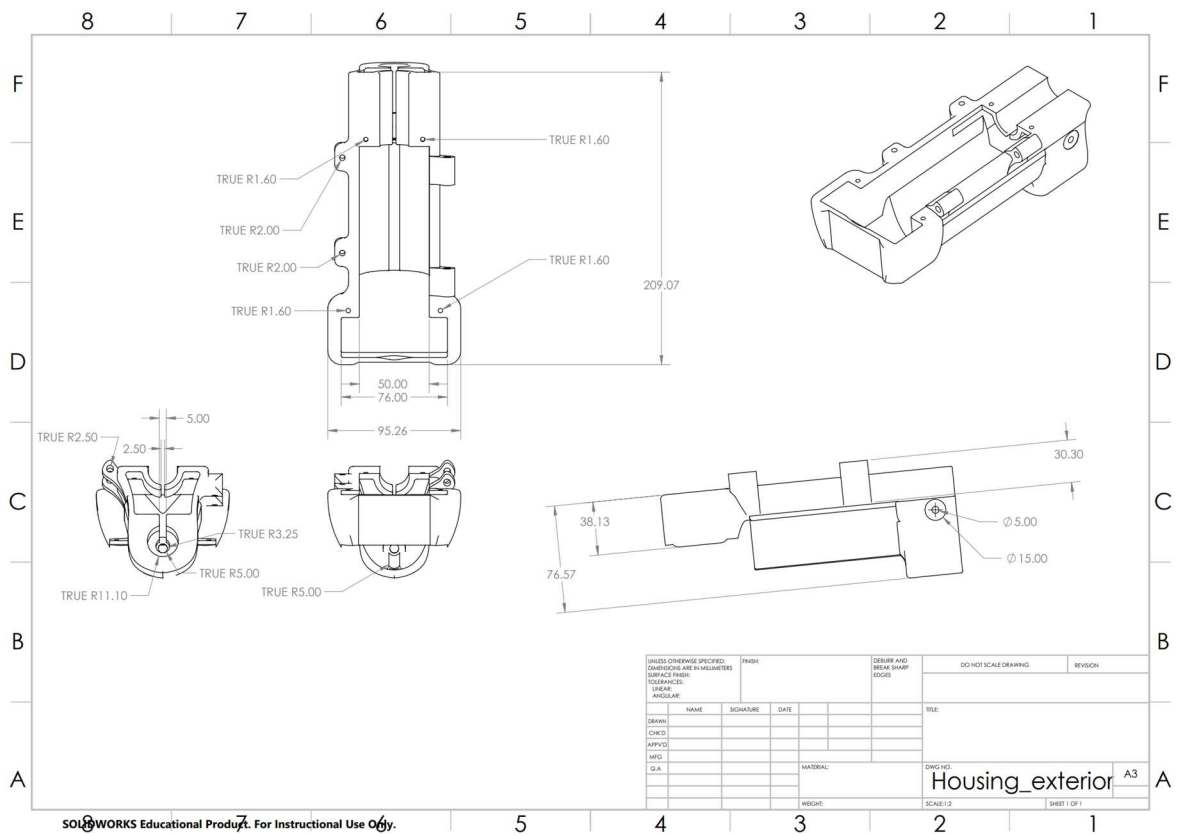


Figure C3: v0.6 housing exterior part drawing

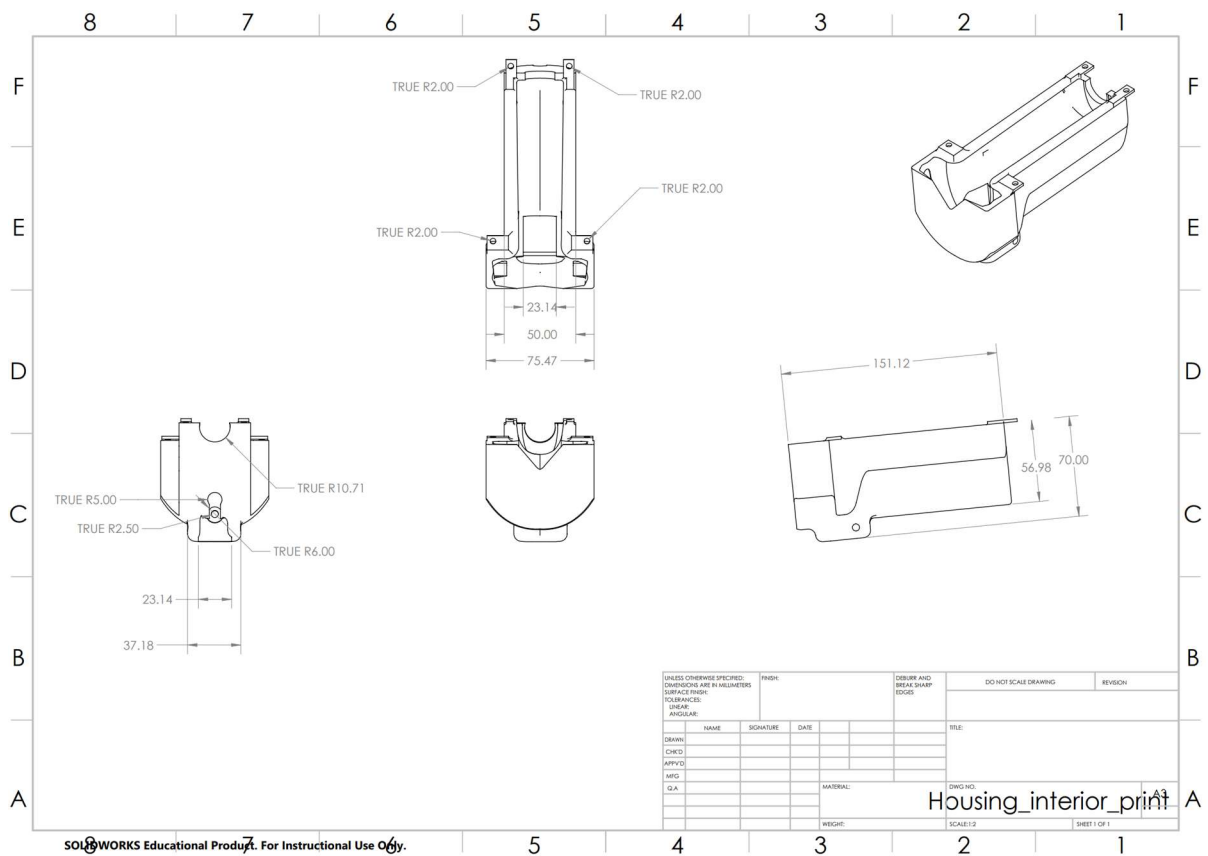


Figure C4: v0.6 3d printed housing interior part drawing

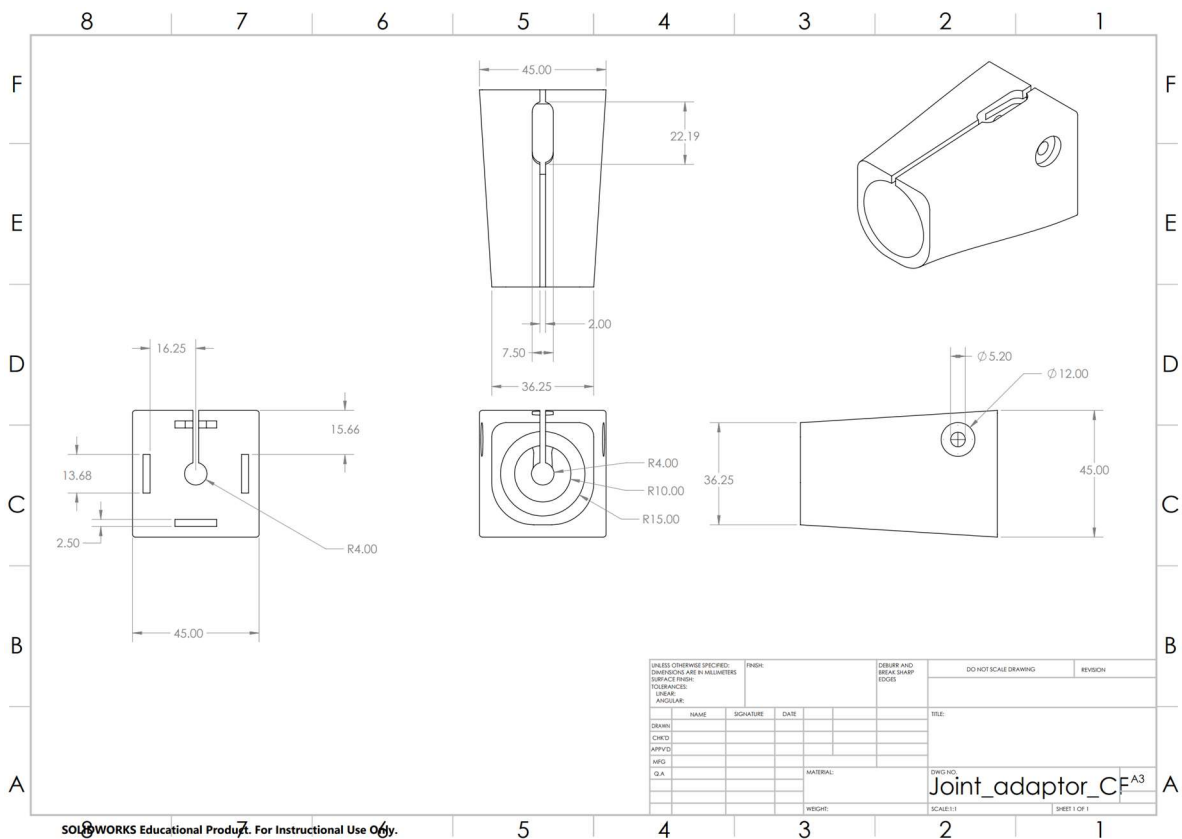


Figure C5: v0.6 Carbon fibre shaft to locking joint adaptor part drawing

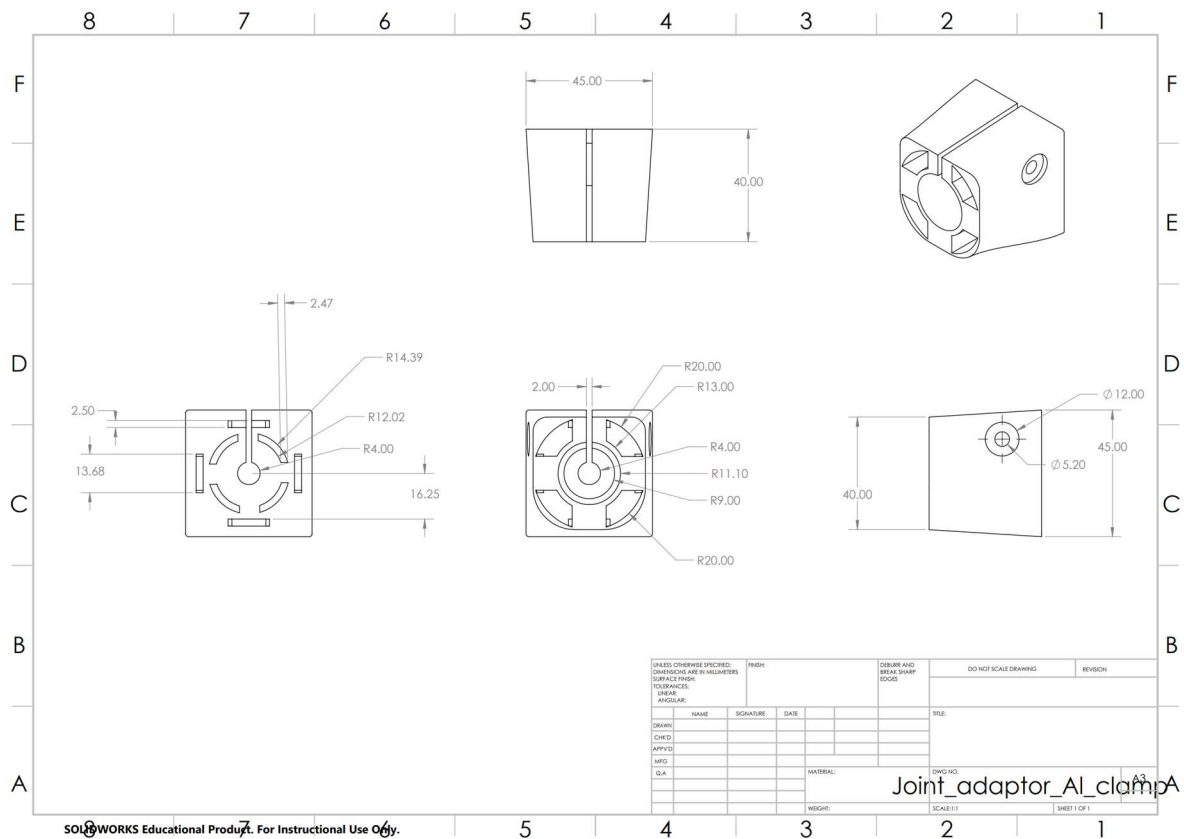


Figure C6: v0.6 locking joint to aluminium housing shaft adaptor part drawing

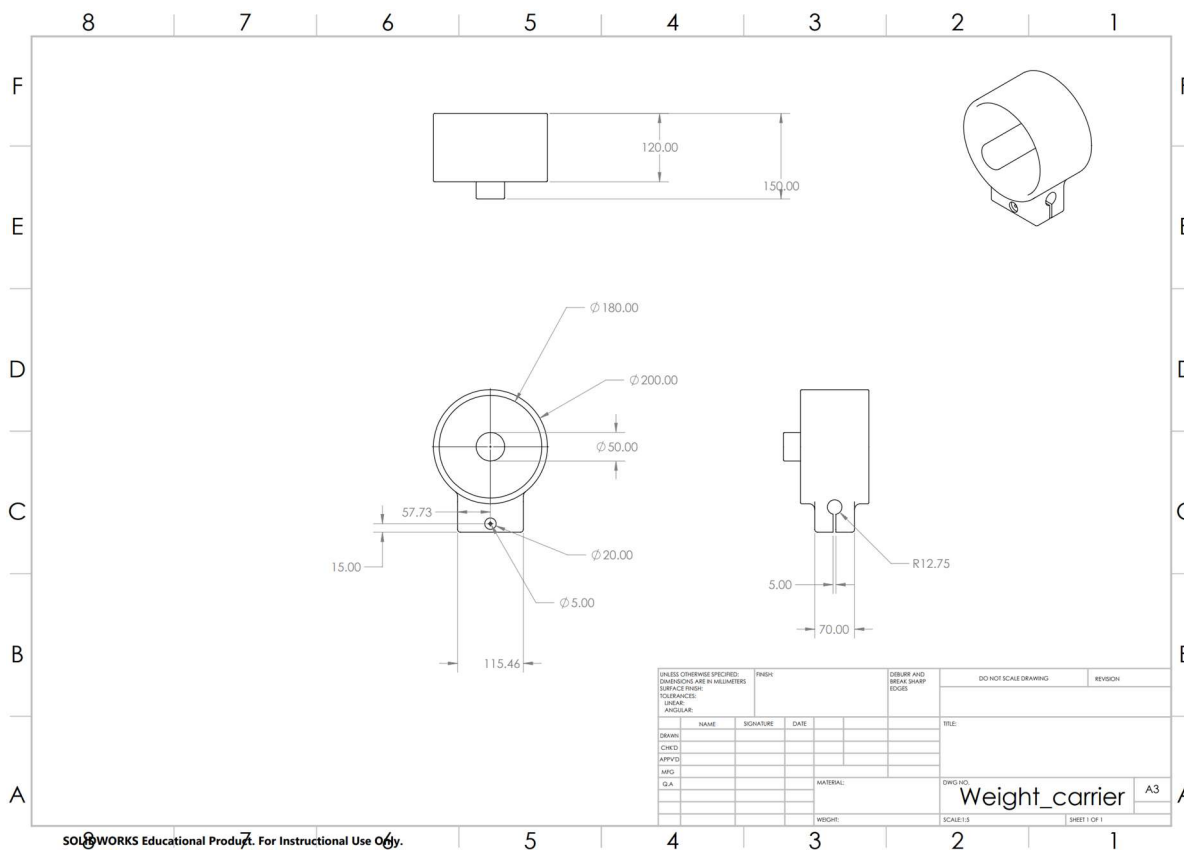


Figure C7: v0.6 weight carrier part drawing

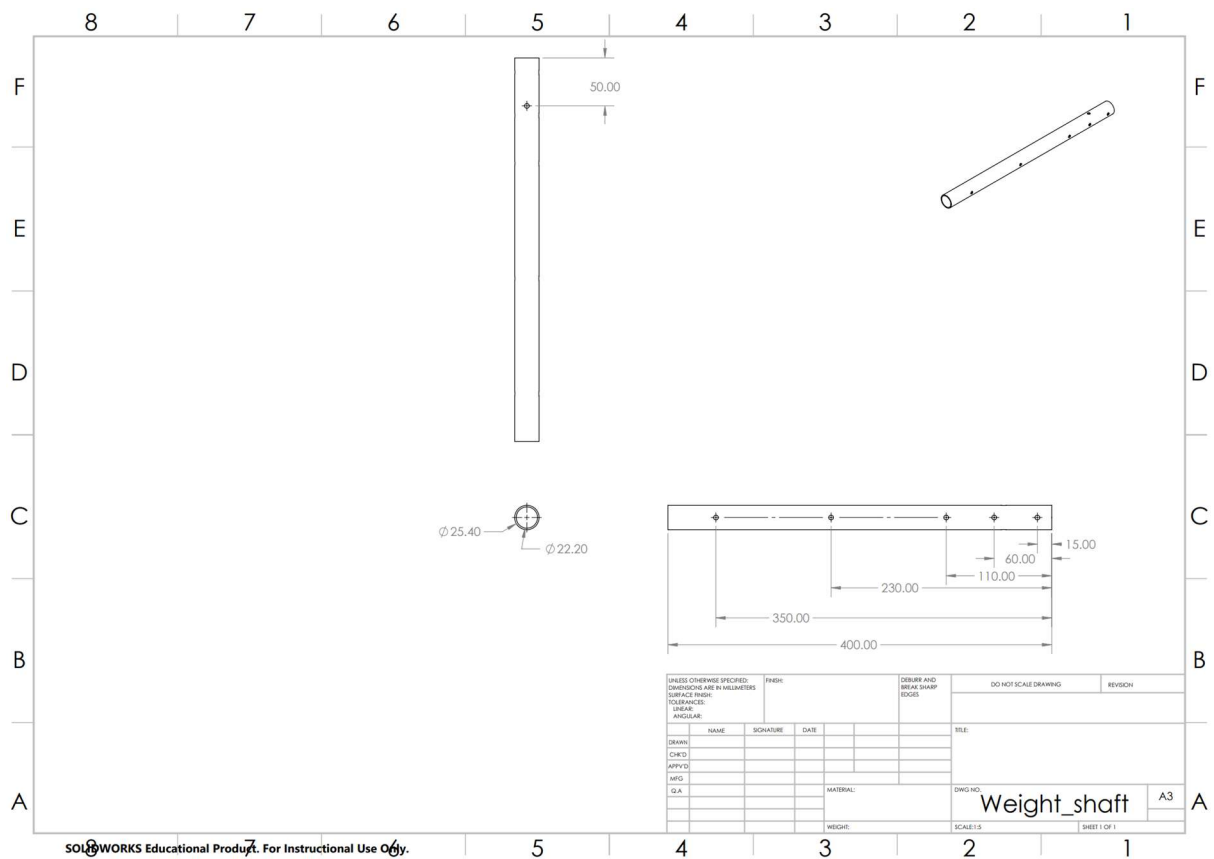


Figure C8: v0.6 counterweight shaft

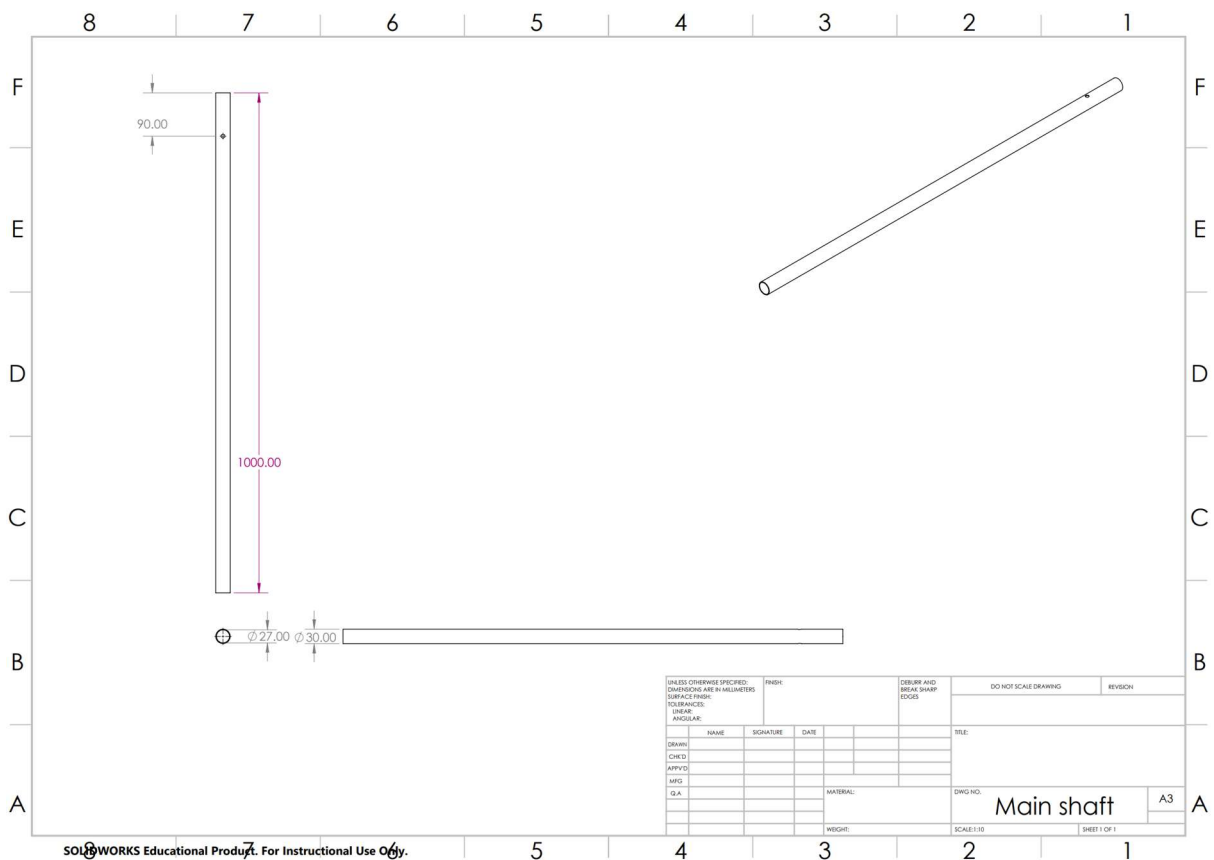


Figure C9: v0.6 main shaft

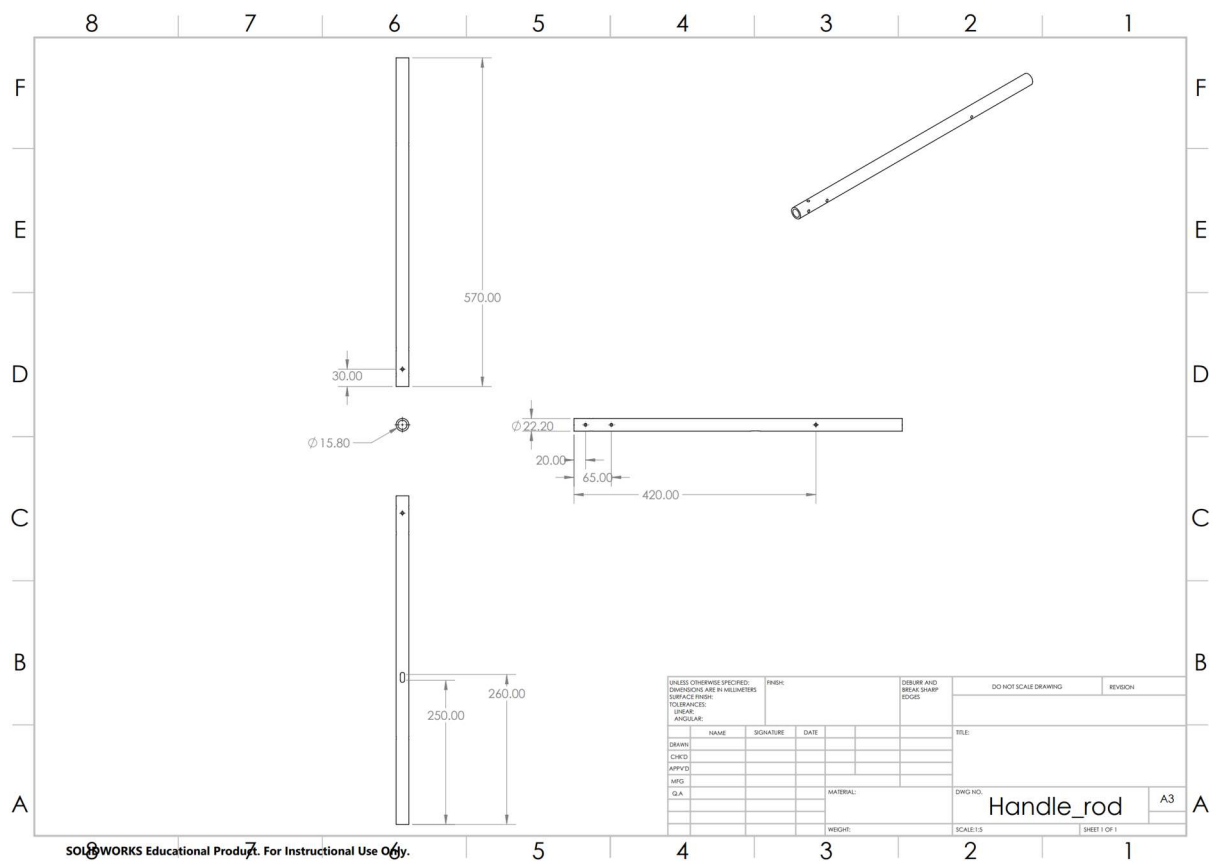


Figure C10: v0.6 handle shaft

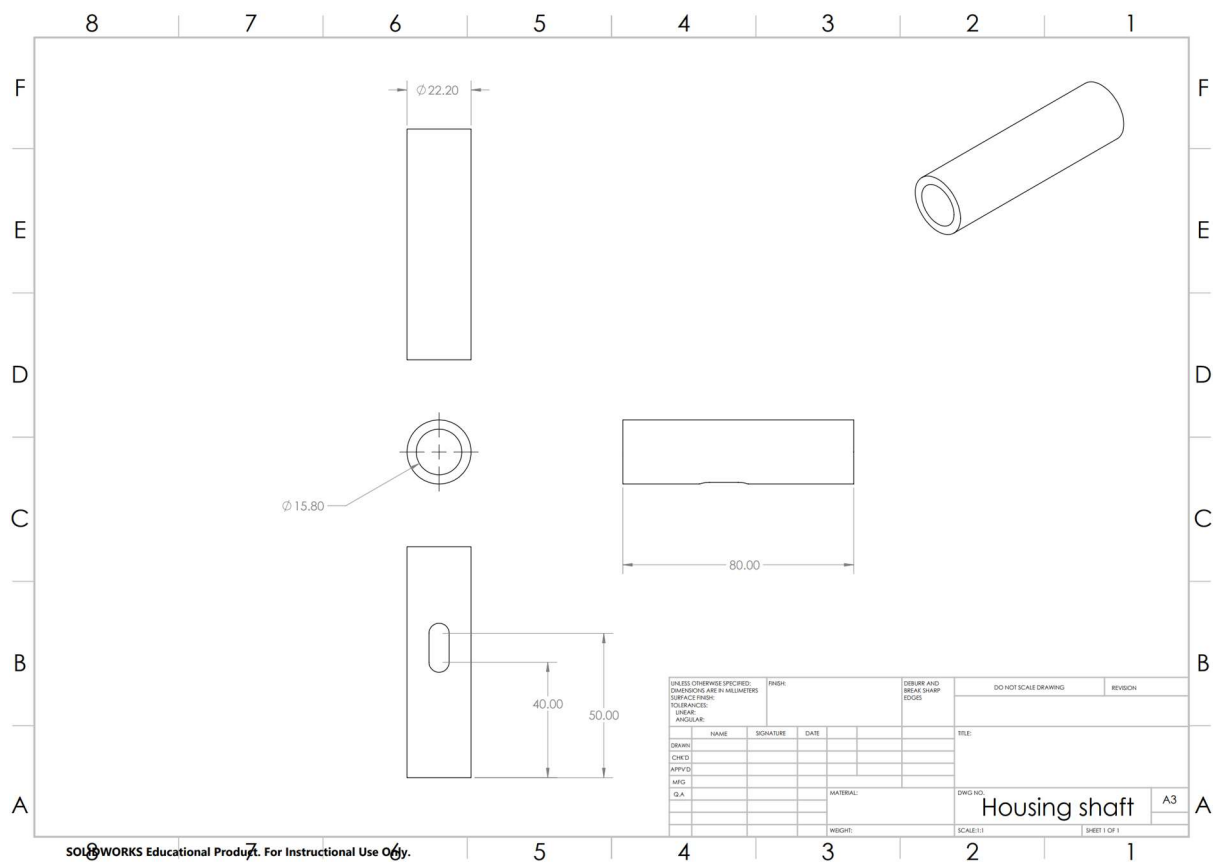


Figure C11: v0.6 housing shaft

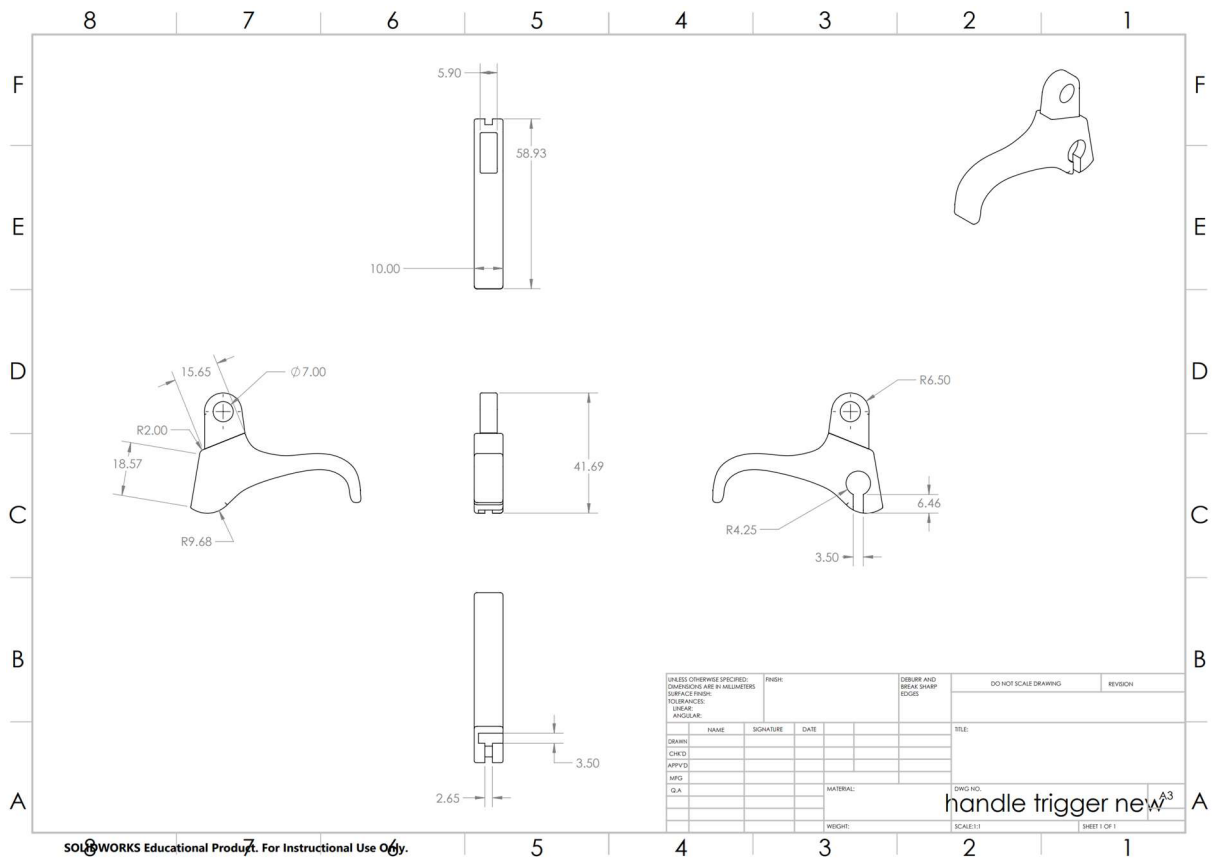


Figure C12: v0.6 custom clutch lever/trigger

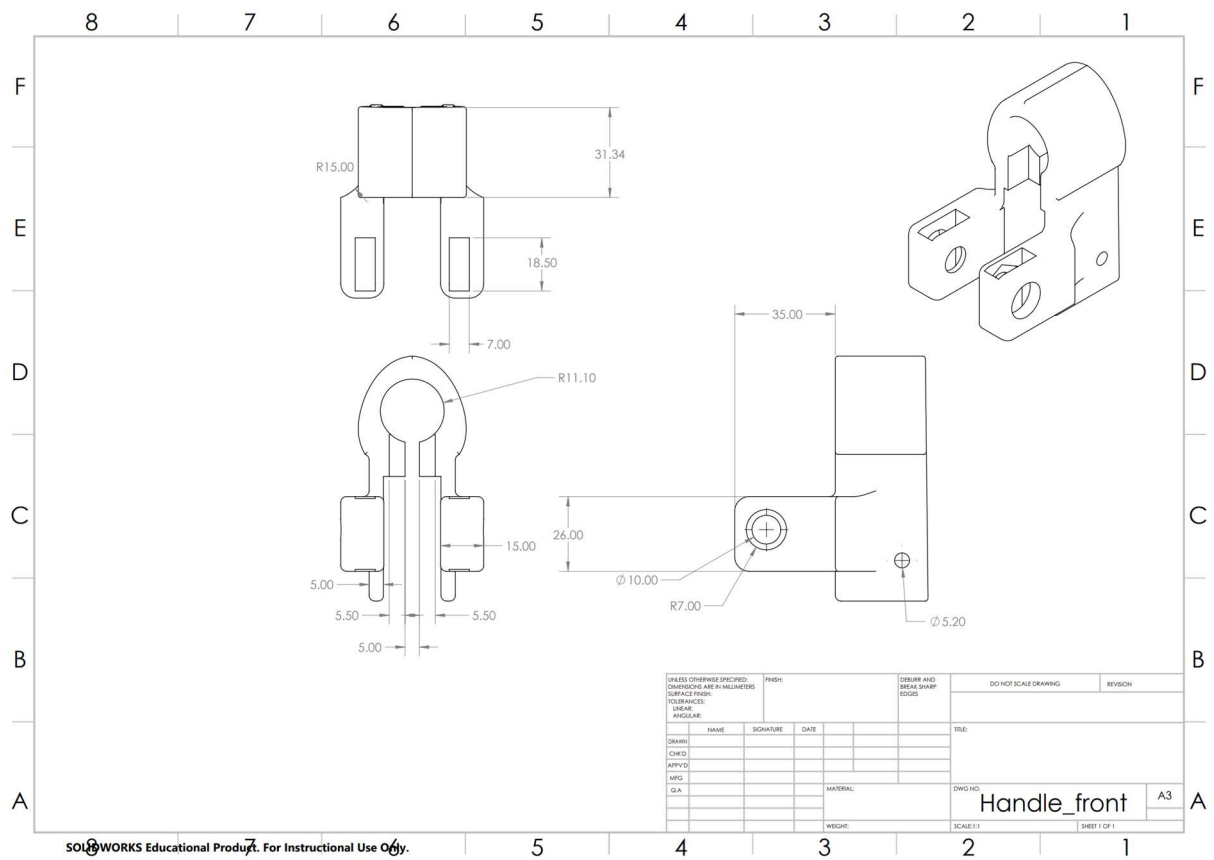


Figure C13: v0.6 front handle attachment

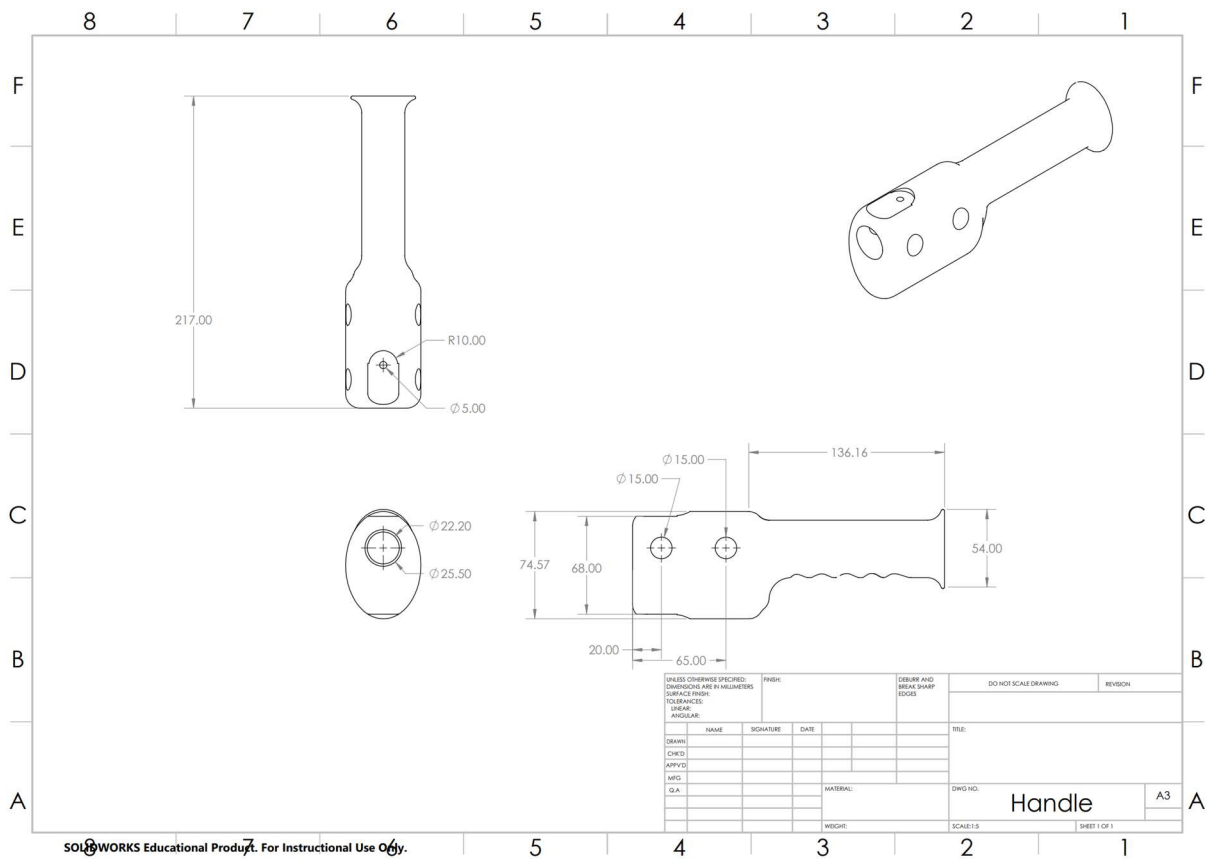


Figure C14: v0.6 handle grip

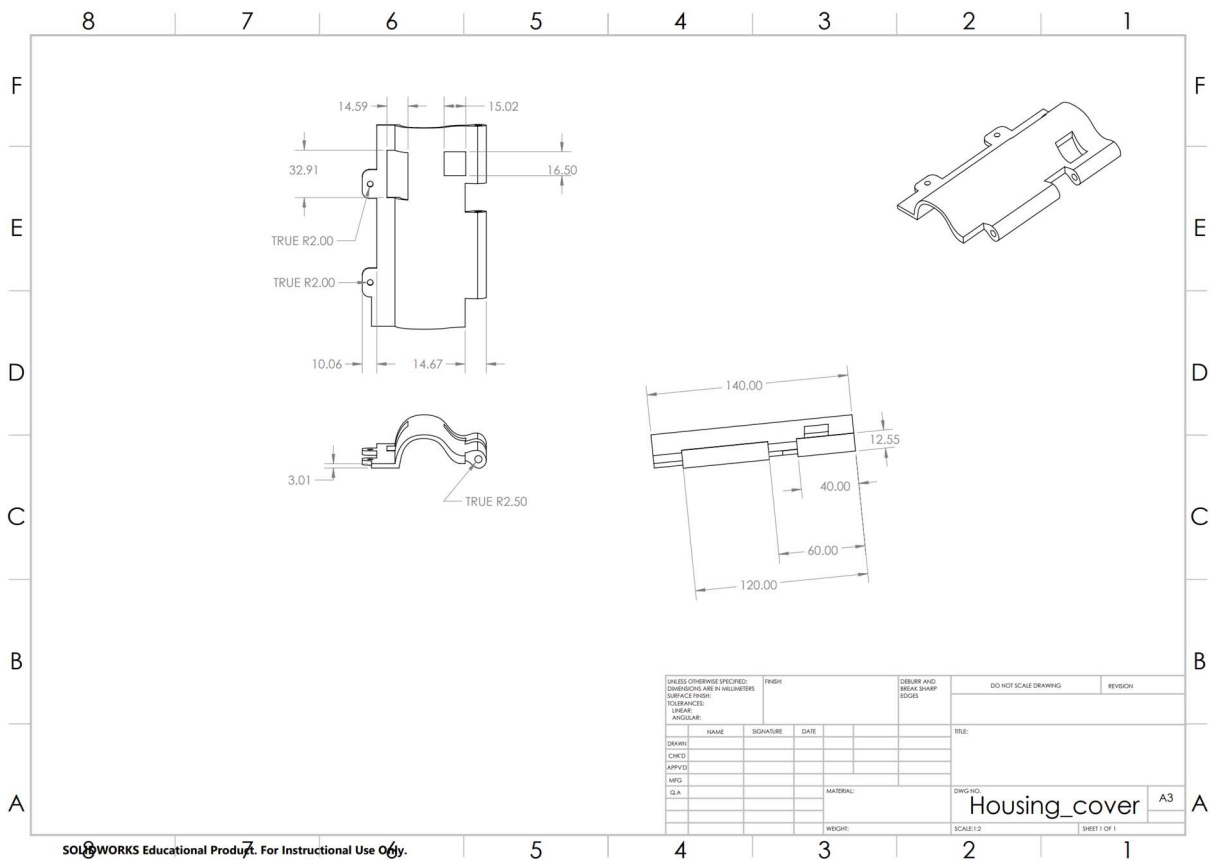


Figure C15: v0.6 housing securing hatch

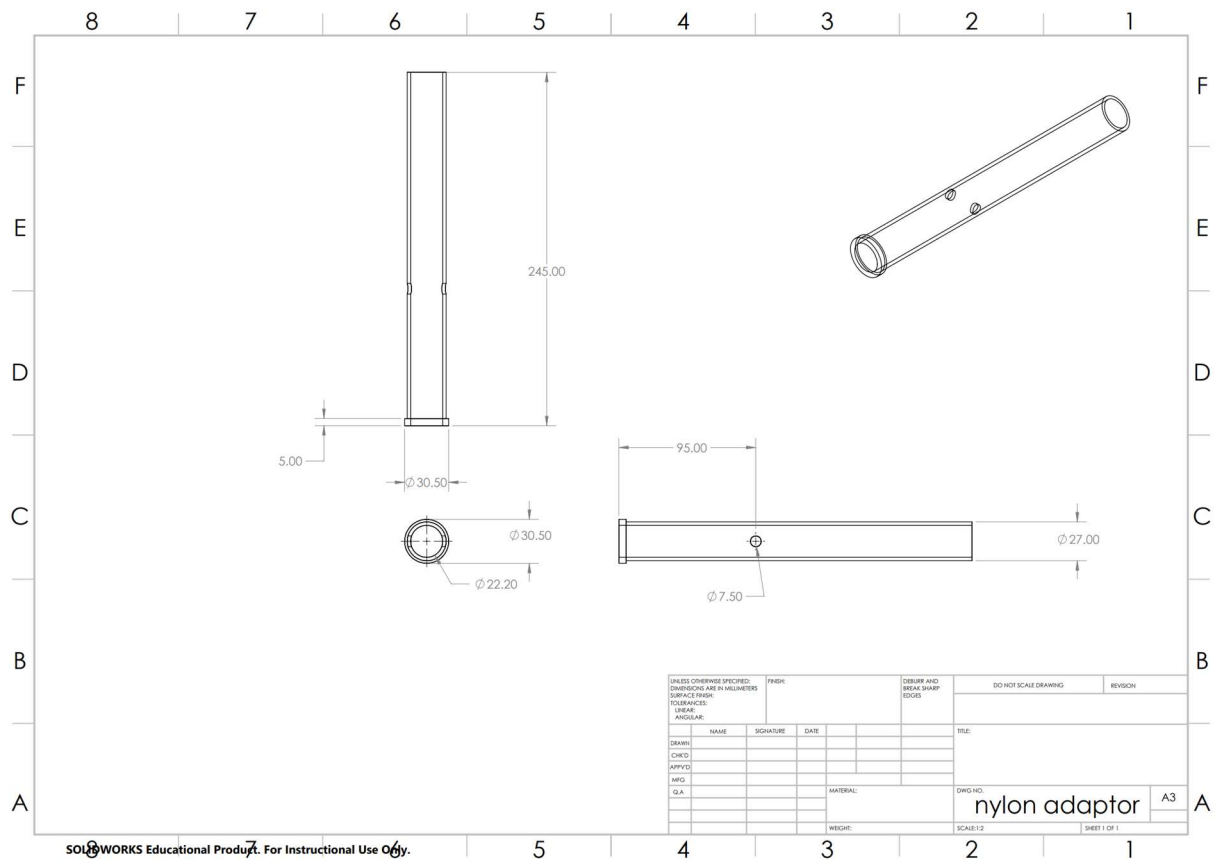


Figure C16: v0.6 nylon 6 shaft adaptor sleeve

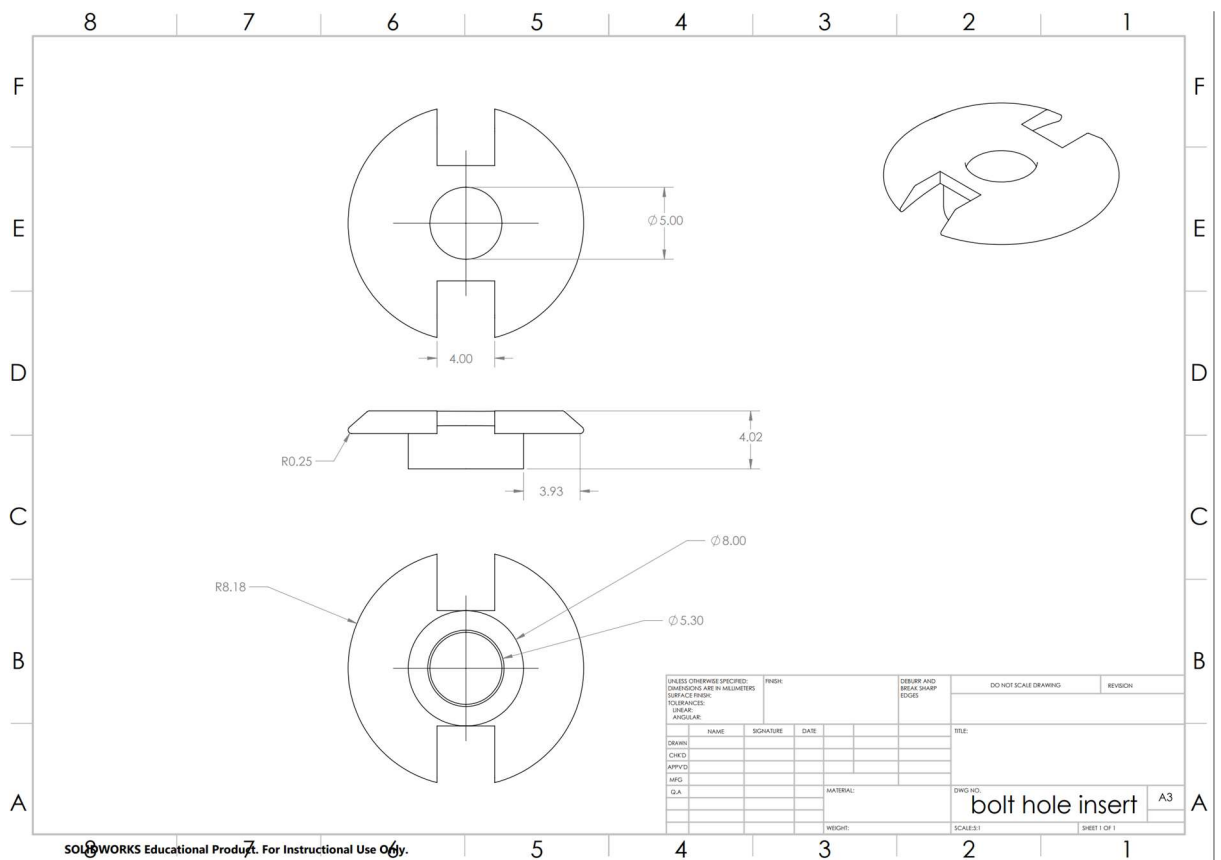


Figure C17: v0.6 protective bolt hole insert

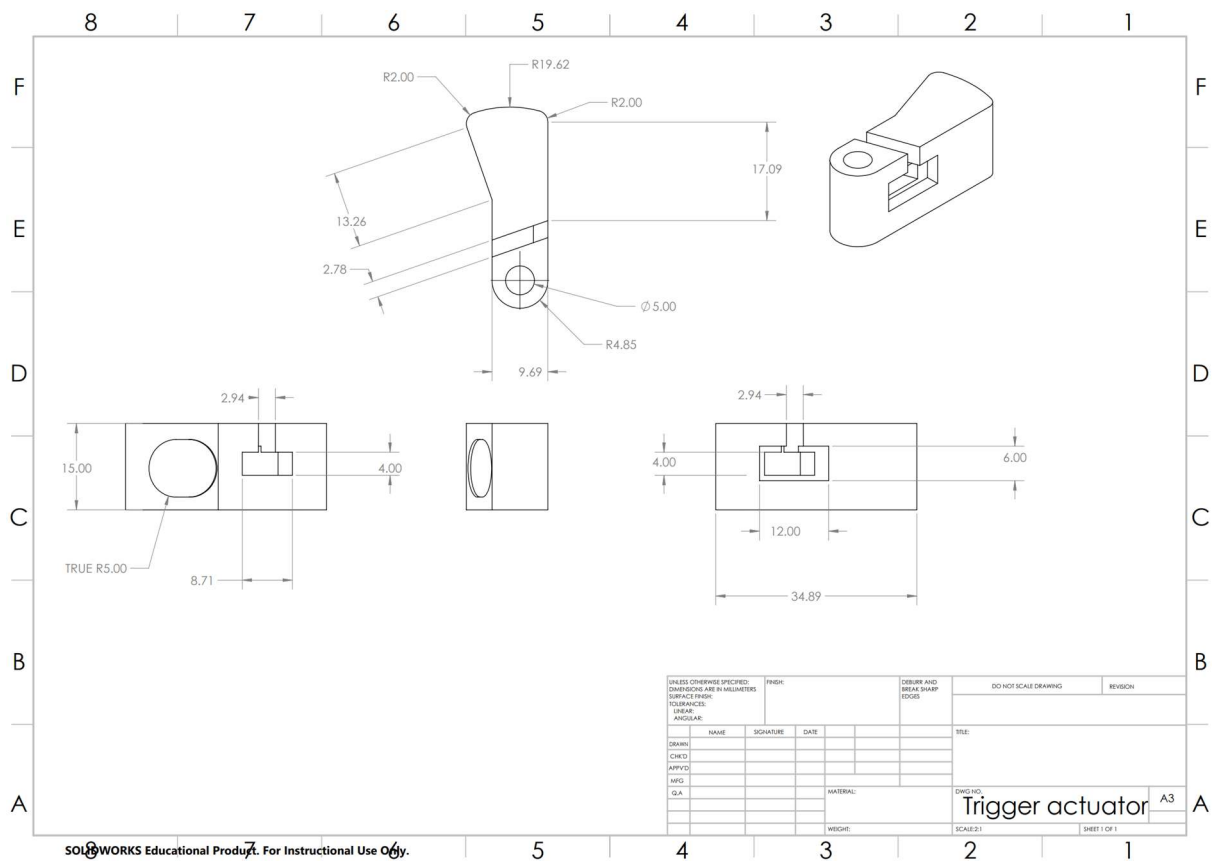


Figure C18: v0.6 trigger actuator

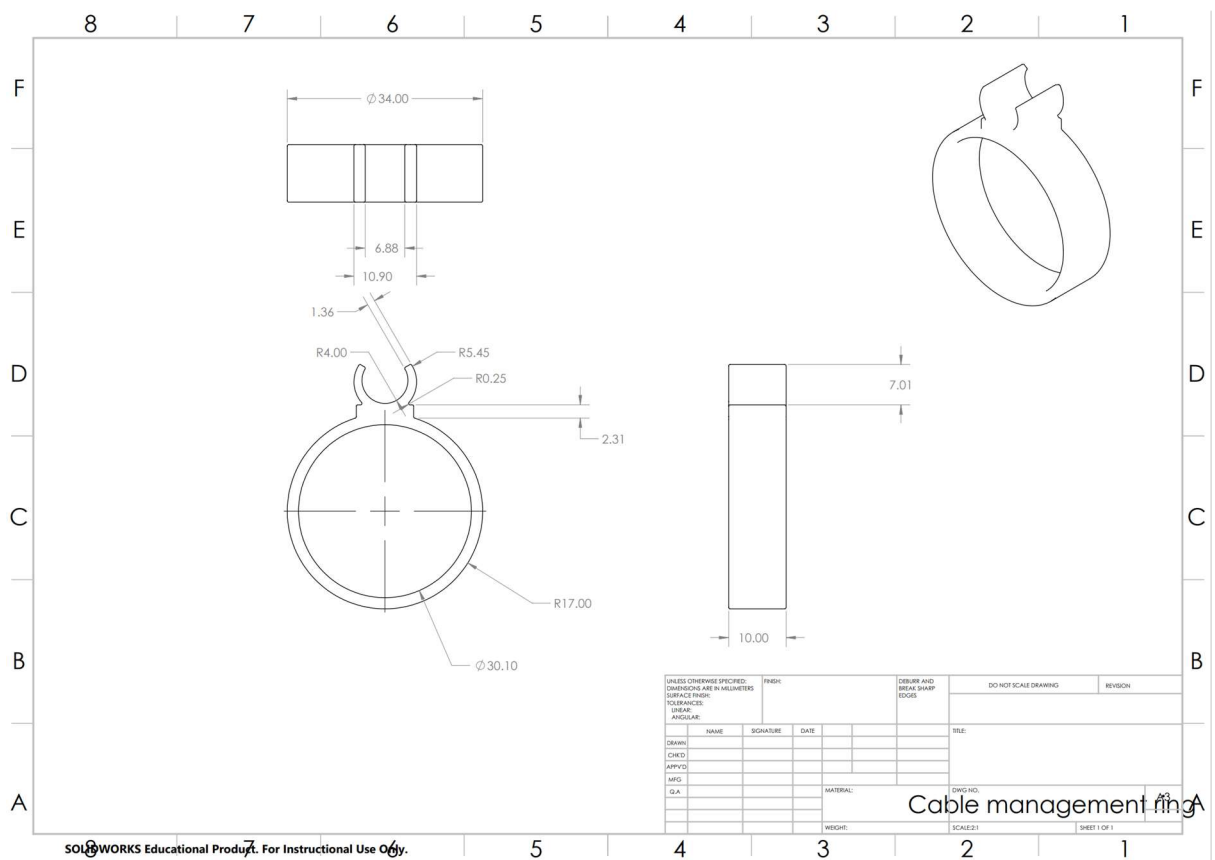


Figure C19: v0.6 cable management ring

Appendix D: Optimisation Program Full Source Code

```

from re import I
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize
from matplotlib.patches import Patch
# Data
#-----
Load = 343 # Total substitute load in N
Torque = 8 # Torque in Nm
Length = 0.08 # Shaft length in m
sigma_a = 340e6 # Max allowable tensile stress in Pa
tau_a = 210e6 # Max allowable shear stress in Pa
t_min = 0.0001 # Minimum thickness in m
t_max = 0.004 # Maximum thickness in m
# Objective function: Minimize the cross-sectional area
#-----
def area_obj(x):
    Ro, Ri = x
    return np.pi * (Ro**2 - Ri**2)
# Functions to obtain stresses / shaft properties
#-----
def inertia(Ro, Ri):
    # Avoid zero / negative inertia that causes divisions by zero or NaNs.
    # clamp = max(Ro^4 - Ri^4, eps)
    eps = 1e-12
    denom = Ro**4 - Ri**4
    denom = np.maximum(denom, eps)
    return (np.pi / 4) * denom
def bending_stress(Ro, Ri):
    I = inertia(Ro, Ri)
    return (Load * Length * Ro) / I
def shear_stress(Ro, Ri):
    I = inertia(Ro, Ri)
    return Load * (Ro**2 + Ro*Ri + Ri**2) / (3.0 * I)
def wall_thickness(Ro, Ri):
    return Ro - Ri
def torsional_shear(Ro, Ri):
    J = (np.pi / 2) * (Ro**4 - Ri**4)
    return Torque * Ro / J
# Constraint functions
#-----
def c1_bending_stress(x):
    Ro, Ri = x
    return sigma_a - bending_stress(Ro, Ri)
def c2_shear_stress(x):
    Ro, Ri = x
    return tau_a - shear_stress(Ro, Ri)
def c3_min_thickness(x):
    Ro, Ri = x
    return wall_thickness(Ro, Ri) - t_min
def c4_max_thickness(x):
    Ro, Ri = x
    return t_max - wall_thickness(Ro, Ri)
def c5_torsional_shear(x):
    Ro, Ri = x
    return tau_a - torsional_shear(Ro, Ri)
# Constraints for optimize function
#-----
constraints = [{'type': 'ineq', 'fun': c1_bending_stress},
               {'type': 'ineq', 'fun': c2_shear_stress},
               {'type': 'ineq', 'fun': c3_min_thickness},
               {'type': 'ineq', 'fun': c4_max_thickness},
               {'type': 'ineq', 'fun': c5_torsional_shear},
               ]
bounds = [(0.0111, 0.0111), (0.005, 0.0111)] # Bounds for Ro and Ri
initial_guess = np.array([0.01, 0.006]) # Initial guess for Ro and Ri
# Optimization
#-----
result = minimize(area_obj, initial_guess, method='SLSQP', constraints=constraints, bounds=bounds)
# Always report status so you can see why it "failed"
print("\ntrust-constr result:")
print(" success:", result.success)
print(" message:", result.message)
print(" status:", getattr(result, 'status', None))
print(" fun (objective):", getattr(result, 'fun', None))
print(" x:", result.x)
Ro_opt, Ri_opt = result.x # defined even if not successful (may contain NaNs if evaluation failed)
if result.success:
    print(f"Optimal Outer Radius (Ro): {Ro_opt:.4f} m")
    print(f"Optimal Inner Radius (Ri): {Ri_opt:.4f} m")
    print(f"Minimum Cross-sectional Area: {area_obj(result.x):.6f} m^2")
# Graphical representation of the constraints and optimal solution
#-----
Ro_vals = np.linspace(0, 0.03, 500)
Ri_vals = np.linspace(0, 0.03, 500)
Ro_grid, Ri_grid = np.meshgrid(Ro_vals, Ri_vals)
I_grid = inertia(Ro_grid, Ri_grid)
sigma_grid = bending_stress(Ro_grid, Ri_grid)
tau_grid = shear_stress(Ro_grid, Ri_grid)
torsion_grid = torsional_shear(Ro_grid, Ri_grid)
feasible = ((sigma_grid <= sigma_a) &
            (tau_grid <= tau_a) &
            (wall_thickness(Ro_grid, Ri_grid) >= t_min) &
            (wall_thickness(Ro_grid, Ri_grid) <= t_max) &
            (torsion_grid <= tau_a))
area_grid = np.pi * (Ro_grid**2 - Ri_grid**2)
area_feasible = np.where(feasible, area_grid, np.nan)
# Plotting
plt.figure(figsize=(10, 6))
# Feasible region (convert bool to numeric for contourf)
plt.contourf(Ro_grid, Ri_grid, feasible.astype(int), levels=[0.5, 1], colors=["lightgrey"], alpha=0.8)
# Constraint lines (normalized; contour at 1)
plt.contour(Ro_grid, Ri_grid, sigma_grid / sigma_a, levels=[1], colors='red', linewidths=2)
plt.contour(Ro_grid, Ri_grid, tau_grid / tau_a, levels=[1], colors='blue', linewidths=2)
plt.contour(Ro_grid, Ri_grid, Ro_grid - Ri_grid, levels=[t_min], colors='green', linewidths=2)
plt.contour(Ro_grid, Ri_grid, Ro_grid - Ri_grid, levels=[t_max], colors='green', linewidths=2, linestyle='dashed')
plt.contour(Ro_grid, Ri_grid, torsion_grid / tau_a, levels=[1], colors='purple', linewidths=2)
# Beam area contours
contours = plt.contour(Ro_grid, Ri_grid, area_feasible, levels=2, colors='grey', linestyle='dashed', alpha=0.8)
plt.clabel(contours, inline=True, fontsize=8)
# Optimal solution (plot only if finite)
if np.all(np.isfinite([Ro_opt, Ri_opt])):
    plt.plot(Ro_opt, Ri_opt, 'ko', marker='*', markersize=10)
plt.xlabel(r"Outer Radius $R_o$ (m)", fontsize=12)
plt.ylabel(r"Inner Radius $R_i$ (m)", fontsize=12)
plt.title("Feasible Region and Optimal Solution for Shaft Design", fontsize=14)

```

```

# Build legend including a patch for the feasible region
feasible_patch = Patch(facecolor='lightgrey', edgecolor='none', alpha=0.8)
legend_lines = [
    plt.Line2D([0], [0], color='red', lw=2),
    plt.Line2D([0], [0], color='blue', lw=2),
    plt.Line2D([0], [0], color='green', lw=2),
    plt.Line2D([0], [0], color='green', lw=2, linestyle='dashed'),
    feasible_patch,
    plt.Line2D([0], [0], color='purple', lw=2),
    plt.Line2D([0], [0], marker='*', color='black', linestyle='none'),
]
legend_labels = [
    "Bending stress limit",
    "Bending shear limit",
    "Minimum wall thickness",
    "Maximum wall thickness",
    "Feasible region",
    "Torsional shear limit",
    "Optimal solution ('SLSQP' method)",
]
plt.legend(legend_lines, legend_labels, loc="upper left")
# Add a small padding so constraint lines are not clipped at the edges
ax = plt.gca()
x_min, x_max = Ro_grid.min(), Ro_grid.max()
y_min, y_max = Ri_grid.min(), Ri_grid.max()
pad_x = 0.03 * (x_max - x_min) # 3% horizontal padding
pad_y = 0.03 * (y_max - y_min) # 3% vertical padding
ax.set_xlim(x_min - pad_x, x_max + pad_x)
ax.set_ylim(y_min - pad_y, y_max + pad_y)
plt.grid(True)
plt.show()
# Printout of final numbers (safe even if optimizer failed)
if np.all(np.isfinite([Ro_opt, Ri_opt])):
    print(f" Ro = {Ro_opt:.9f} m ({Ro_opt*1000:.6f} mm) => External Diameter = {2*Ro_opt*1000:.6f} mm")
    print(f" Ri = {Ri_opt:.9f} m ({Ri_opt*1000:.6f} mm)")
    print(f" t = {Ro_opt-Ri_opt:.9f} m ({(Ro_opt-Ri_opt)*1000:.6f} mm)")
    print(f" Area = {area_obj(result.x):.9f} m^2 ({area_obj(result.x)*1e6:.6f} mm^2)")
else:
    print(" Final x contains non-finite values; optimization encountered numerical problems.")

```

Appendix E: Project Gantt Chart

Table E1: Project Gantt Chart

[illegible]

Appendix F: Supervisor Meeting Record

SCHOOL OF ENGINEERING, CARDIFF UNIVERSITY

Department of Mechanical and Medical Engineering

2025-26

EN3100 - Project Meeting Attendance Sheet

NAME:Luke Lewis-Atwell.....**STUDENT NUMBER:**23056446.....

SUPERVISOR:Dr Alastair Clarke.....

PROJECT TITLE:Design of an extended handle for a laser scanning tool.....

TEACHING DISCIPLINE: Mechanical OR Medical Engineering?Mechanical.....

Date of meeting On-line or F2F?	Student's summary of progress since last meeting – to be completed before the next meeting	Supervisor's assessment of progress	Actions by next meeting
02/10/2025 F2F	Initial meeting	Good start!	Start recording research around similar solutions and start developing a project timeline.
09/10/2025 F2F	Started schedule and found some relevant existing designs	Discussed meeting with company. Good background research.	Prepare questions for hexagon on Mon. Look for tech specs from company. Start working on presentation draft
16/10/2025 F2F	Discussed the CAD models from MES and looked at a draft of wk4 presentation	Good.	Finish presentation, ergonomics research and thinking about the start of the design process
24/10/2025 F2F	Finished and delivered presentation, came up with reflector concept	Presentation prepared – well done.	Sketch some concepts for MES to discuss, continue looking at ergo and each required feature. <u>Start Phase 2</u>
30/10/25	Showed initial sketches and explored some design concepts	Good initial designs.	Turn sketches into more complete concepts
10/11/25	Showed Prepared concepts for MES feedback	Good progress.	Contact MES asap for feedback before starting CAD
13/11/25	Relayed approval of the concepts and slight changes from MES. Demonstrated fibre optic working for light pickup	Excellent work.	Start modelling. Look again for more accessible hinge.
20/11/2025	Discussed progress and issues with CAD. Showed printed testing grip. Reminder of progress review week 10. Looked at joints instead of hinges	Steady progress with CAD.	Further scouting for joints/ other connector designations. Continue with modelling
27/11/2025	Showed CAD model assembly of major components, discussed issues with assembling current handle	Good to see CAD assembly progressing.	This week: Focus on progress review. Next week: continue modelling and focus on details and part selection. No meeting planned for 4/12/2025.
11/12/2025	Discussed the need for a cable guide to avoid slacking in the bowden cable and redesigns of the counterweight and handle. Also some details need further refinement (fastener locations and trigger actuator)	Good progress.	Refine details and aim to start FEA as soon as possible after exams.

04/02/2026	Showed current state of model and initial import of assembly into ansys. Discussed procurement of plunger, joint and bowden cable. Discussed Aluminium vs Carbon fibre shafts (Easy composites)	Good progress.	Change ansys approach to part by part, calculating moments from the handle for applied forces. Also, do hand calculations for shaft bending to find rough cross-section requirement. Consider clamp style redesign for shaft attachments. Create a parts list ready for procurement.
11/02/2026	Discussed issues with FEA on 3d printed parts, showed optimisation program for shaft cross-section and some parts for procurement. Also discussed new solution using aluminium core/supports and 3d printing non-structural parts with more complex geometry	Detailed discussion on materials choice. Good progress.	Send parts list by end of week, redesign for Al + 3d print structure. Change shaft securing method to clamp style. Put more work into report
25/02/2026	Showed v0.3 redesign using aluminium structure and 3d prints. Shaft sizes on procurement list validated with optimisation code. Discussed lack of progress with documentation.	Great work on re-design.	FEA still needs to be completed. Documentations needs much more work generally, and for draft chapter feedback around middle of week 7
04/03/2026	Discussed challenges with different FEA approaches attempted and likely stress singularities in the simulations.	Good to see detailed FEA	Finish FEA by next week. Refine documentation from notes to a draft.
11/03/2026	Received several parts (joint, CF shaft) - only aluminium shafts still undelivered. Demonstrated V.5 cad model. Completed FEA on version 5 - changed to full aluminium shaft due to concerning deflection of CF shaft. Discussed deadline extension due to computer resource limitations and procurement delays.	Great progress - nearly done!	Do physical testing of Carbon fibre shaft. If it is suitable revert to CF shaft for final version. Produce final part drawings and end the project. Fully pivot to documentation to produce a sample draft chapter for wed 18/03